Aligned but Apart: Why Cosine Conflict Diagnostics Misread Auxiliary-Loss Interference at the Output Projection

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Abstract

Cosine similarity between task gradients is the standard instrument for detecting *conflict* in auxiliary-loss training: a negative cosine reads as interference, and gradient surgery acts on it. At a language model's shared, tied output projection it is misleading. We measure the per-row gradients a next-token (CE) and an auxiliary multi-token (MTP) loss deposit there, and find them aligned in direction. On the ≈8% of rows that receive a target the median per-row cosine is ≈0.53 and only ≈3–4% are opposed; the remaining ≈92% are scalar multiples, cosine +1 by construction. The tensor is *tied*, so an embedding-path contribution to those cosines is unbounded here (Sections 3.2, 6). Yet the *aggregate* cosine runs from −0.08 to −0.19 under Muon, shallow enough at one end to read as benign, and −0.28 to −0.34 under AdamW. What comes apart is the two readings, not the supports: per-row gradient *norms* (three seeds, both optimizers) put both losses on the *same* rows, at a norm-profile cosine of 0.98, but 60–69% of that mass sits on ≈0.3% of rows, the most frequent function words and punctuation. This is norm-weighted cancellation, not disjoint support. An exact rewrite shows the shared-head objective is cross-entropy toward a next-and-future-token mixture, whose zero-parameter KL cost reproduces 62–91% of the matched within-sweep gap across estimator variants. That makes the shifted optimum, not gradient conflict, the leading account of the +0.39-nat degradation we measure at 124M ($p{\approx}8{\times}10^{-5}$). Here the aggregate cosine is *uninformative about where interference lives.*

1 Introduction

Predicting several future tokens at once (multi-token prediction, MTP) has become a common ingredient in large-scale language modeling, used to improve sample efficiency and to enable self-speculative decoding (Gloeckle et al., 2024; DeepSeek-AI, 2024). The auxiliary objective is double-edged: a growing body of work reports that adding it can *degrade* the primary next-token objective, particularly at smaller scales and longer horizons (Gloeckle et al., 2024; Zuhri et al., 2025). Asked why, practitioners reach for the intuition that pervades multi-task learning: the two losses must pull the shared parameters in *conflicting directions*, and the remedy is to detect and neutralize that conflict, typically from the sign of a gradient cosine (Yu et al., 2020; Wang et al., 2021). This intuition has been questioned in auxiliary-task learning: Jiang et al. (2023) report that gradient-conflict coordination can overlook auxiliary-target generalization, so the cosine signal does not by itself determine negative transfer. We ask what that signal actually measures at a language model's shared output head.

This paper is a measurement study of that intuition for the case of an auxiliary multi-token objective sharing a model's tied output projection. Our central finding is diagnostic: **the cosine conflict signal is misleading here.** We instrument the gradient each loss deposits on the output projection and examine it **row by row** (one row per vocabulary item). Two facts emerge, both from data already in hand. First, the per-row gradients are not opposed; they are *aligned*. We measure this on the rows that receive a target in the measurement batch, the ≈8% that are not scalar multiples by construction (Section 4.1). On those rows the median per-row cosine is ≈0.53, and ≈3–4% are opposed. Over all rows the median is +1 and 98.4% have $|\cos| > 0.3$. On the matched active-row denominator, 83% of active rows have $\cos > 0.3$; a control that measures CE against an unrelated L1 loss on the same rows reads 1.0%. Either way the per-row directions are predominantly aligned, in sharp contrast to the near-zero aggregate. Second, the *aggregate* cosine over the flattened matrix is nonetheless near zero, and tracking it over training shows the two metrics **start together** (both $\approx +0.99$ at initialization) and then **decouple**: within $\sim$200 steps the aggregate falls below zero while the per-row median stays at $\approx +1$. By count, almost no rows are opposed. But the aggregate cosine is magnitude-weighted, so we instrumented the per-row gradient norms directly to see where the weight lies. The two losses load their magnitude on the *same* rows: the norm-profile cosine, meaning the cosine of the two per-row norm vectors, is 0.98. Yet a tiny high-norm opposed minority, $\approx$0.3% of rows, carries 60–69% of the mass across optimizers. It is this minority that pulls the flattened sum negative. The alignment is real and holds for the vast majority of rows; the aggregate is a *norm-weighted cancellation* the cosine cannot localize.

We frame this norm-weighted opposition as the structure a cosine diagnostic cannot see (Figure 1), and we relate it to the observed degradation without claiming to have isolated its cause. Adding the shared MTP objective degrades next-token loss by $\approx$0.39 nats ($\approx$10%) at 124M, a large and unambiguous effect (Welch's unequal-variance t-test, $p \approx 8 \times 10^{-5}$), yet it co-occurs with aligned, not conflicting, per-row gradients (the degradation is measured on the full 30,517-step run; per-row alignment on the diagnostic runs above). Three further observations, all from committed runs, are consistent with a norm-weighting/capacity account rather than a directional-conflict one. (i) Conflict-targeted gradient surgery does not recover the baseline even though the opposition it targets is real and mass-dominant: the opposed rows are negligible by count (0.3%) but carry the majority of the gradient-magnitude mass (60–69%), and per-row Gram–Schmidt on the head (Appendix A) removes exactly that opposition, yet the degradation does not move. The opposition the aggregate cosine detects is mass-dominant and causally inert (single-seed; Appendix A). (ii) Detaching the auxiliary's gradient from the shared head makes the damage *worse*, not better. The probe forms the auxiliary logits as $\mathrm{sg}(W)h$, so the auxiliary still trains the trunk and only the head's auxiliary gradient is removed; it is confounded because the head can no longer co-adapt to the representations the auxiliary shapes, and we treat it as suggestive only. (iii) An auxiliary that avoids the shared head (a layer-split contrastive auxiliary) reduces but does *not* eliminate the gap. These auxiliaries change the objective as well as its capacity, so the capacity ingredient is not isolated; the untested comparison that would isolate it is separate-head MTP (Section 6). The figures above are for the contrastive auxiliary: its standard variant is demonstrably short of the CE-only baseline (Δ+0.039, $p \approx 0.010$), while a hyperparameter-tuned variant sits below our detection floor, the smallest difference these seed counts can resolve (Δ+0.018, $p \approx 0.19$), and is inconclusive. We rest the capacity conclusion on the standard variant.

An exact rewrite of the shared-head objective ties these three observations together. The objective is cross-entropy against a next-and-future-token mixture label, in effect label smoothing toward the future (an analogy: conventional label smoothing mixes toward the uniform distribution, whereas this mixes toward future-token targets). Its minimizer is therefore not the next-token distribution, which makes the degradation a plausible cost of the target rather than of the optimization path. On that reading the norm-weighting/capacity observations are subsumed rather than displaced: surgery would be inert because reprojecting gradients does not move an optimum (i; exact for symmetric projection, empirical for our head-only variants, Section 5), and separating capacity would help because it stops the two objectives from sharing one label (iii). A zero-parameter estimate of the implied KL cost accounts, raw, for 62–91% of the matched within-sweep gap (0.3715) across estimator variants; the coverage-extrapolated full-mixture variant lands at $\approx$88–89%, leaving a residual we name rather than assign.

These findings carry a methodological point beyond MTP. Cosine-based conflict detection is standard practice in multi-task and auxiliary-learning optimization (Yu et al., 2020; Wang et al., 2021; Du et al., 2018), and it is the wrong instrument in this regime, misleading in two opposite directions depending on the optimizer. Under Muon the head aggregate sits near zero (-0.08 to -0.19 across measurement batches), which a practitioner treating near-zero as benign reads as "no interference" while the loss degrades regardless. Under AdamW it goes negative enough to trip a conflict flag, yet surgery targeting exactly that opposition recovers nothing. Both readings are of the *head* aggregate; a practitioner applying a strict sign rule, or aggregating over all parameters, would flag conflict under both optimizers (Section 4.4). The decomposition that exposes the problem separates **direction** (cosine) from **support** (where each gradient's magnitude lives). Two additional measurements sharpen the picture and guard against over-reading it. The high per-row alignment is **general** across the network's weight matrices (attention and MLP matrices show it too, many with strongly *negative* aggregate cosine), so the output head is not special because the metric singles it out; it is consequential because it directly produces the logits. And the alignment **persists when the head is untied**, so it is a property of the output-projection gradient, not an artifact of weight tying. A full-vocabulary reading suggests rarer tokens align more, but we show this is a construction artifact: rare tokens are almost never active, so their rows are dominated by the cosine-+1 parallel majority, and on active rows the effect vanishes. Finally, our account is a direction-space, multi-loss complement to Godey & Artzi (2026), who show that backpropagating a *single* loss through the low-rank output head destroys most of its gradient *norm*; we ask what a *cosine* diagnostic reports when *two* losses share that head, and answer that their per-row directions align while a high-norm opposed minority sets the aggregate: measured directly via the per-row gradient norms (Section 4.3), not inferred.

We are explicit about scope. Every result is for a **single shared tied output head**, at 124M (n=5) and 350M (single seed), undertrained, small-batch. We do *not* claim that multi-token prediction *as practiced* (separate per-horizon heads or modules, typically feeding a shared unembedding) degrades next-token modeling; establishing that would require a separate-head experiment we identify as the primary future step (Section 6), along with n=3 at 350M and matched-seed surgery. What we do establish, on data in hand, is that the standard cosine conflict diagnostic misreads this regime, and why.

Contributions. All on a controlled, reproducible GPT-2 testbed (124M primary, 350M illustrative; FineWeb-Edu) with open code and a committed statistics script:

- **A diagnostic result (Figures 2, 3).** For an auxiliary multi-token objective sharing the tied output projection, the per-row gradients of the two losses are *aligned in direction* on the rows that carry a target (active-row median cosine ≈0.53, ≈3–4% opposed; over the full vocabulary the median is +1 because ≈92% of rows are scalar multiples by construction), while the aggregate cosine is ≈ 0, and this decoupling *emerges during training* from an aligned start. Standard cosine-based conflict diagnostics therefore *misdiagnose* this regime.
- **A norm decomposition of the collapse, and a direct measurement of its cause (Figures 4, 6).** Via an exact identity, the aggregate would read +0.94/+0.96 if gradient magnitude were flat across rows but reads −0.08/−0.28: the collapse is set by per-row *norm* structure, not by the typical per-row direction. Instrumenting the per-row norms directly (`measure_norm_support.py`; n=3 seeds per optimizer) settles which structure. The two losses load the *same* rows: norm-profile cosine 0.98. A high-norm opposed minority carries the majority of the mass: opposed-norm fraction 0.60 under Muon and 0.69 under AdamW. The aggregate is a norm-weighted cancellation, not disjoint support.
- **A calibrated, general characterization (Figures 7, 10).** The alignment is **general across the network** and **survives untying** the head; the apparent rarer-tokens-align-more effect is a construction artifact that vanishes on active rows. The next-token degradation is large and significant (+0.39 nats, $p \approx 8 \times 10^{-5}$) yet co-occurs with aligned gradients.
- **Intervention evidence against a directional-conflict account (Figure 9, Appendix A).** Conflict-targeted surgery does not recover the baseline; stop-gradient makes the damage worse; and the standard capacity-separated auxiliary reduces but does not eliminate the gap (the standard variant demonstrably so; the tuned variant inconclusive), each reported with its statistical test and caveats.
- **An account of what the shared head actually optimizes, and a zero-parameter test of it (Figure 8).** An exact rewrite shows the shared-head objective is 1.75× cross-entropy against a next-and-future-token mixture label, so its optimum moves and the next-token cost is bounded by $\mathrm{KL}(P_1 \| q^*) \leq \log 1.75$. Estimating that KL from the CE-only model's own distributions, with no fitted parameters, reproduces the measured degradation across an anchored auxiliary-weight sweep, matching within 0.1σ at the lower weights and falling 1.4–1.8σ short at full weight. This makes the shifted optimum, rather than gradient conflict, the leading account of the loss increase, and it

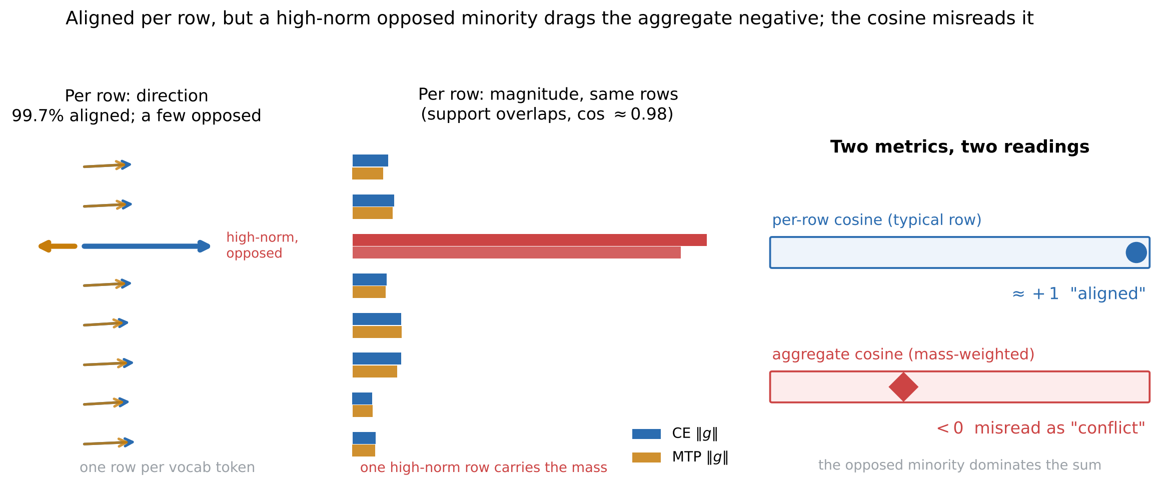


Figure 1: **Most rows agree; a high-norm minority sets the sign** (schematic; illustrative proportions, not plotted data). *Left:* row by row (one row per vocabulary token), the CE and MTP gradients point almost the same way for 99.7% of rows (per-row cosine $\approx +1$); a tiny high-norm minority (red) is opposed. *Middle:* both losses deposit their gradient *magnitude* on the *same* rows (norm-profile cosine ≈ 0.98), and one high-norm row (red) carries most of the mass. *Right:* the typical per-row cosine reports alignment, while the magnitude-weighted aggregate, the standard conflict diagnostic, is dragged negative by the opposed high-norm minority and is misinterpreted as conflict. The aggregate is not wrong about the update, but it cannot say *where* the interference lives: it is set by per-row norms, not by the directions it averages. The 99.7% on the left is the fraction of rows that are *not* opposed, and the $\approx +1$ is dominated by the ≈92% of rows that receive no target in the batch and are exact scalar multiples. On the ≈8% that do carry supervision the median cosine is ≈0.53 with ≈3–4% opposed (Section 4.1); those are the quantities the paper's claims rest on.

explains why conflict-targeted surgery cannot help: symmetric projection leaves a shifted optimum stationary, and the head-only variants we test are empirically inert. We name the residual rather than claim the account is confirmed.

- **A bridge to the output-layer bottleneck literature.** We position the result as the direction-space, multi-loss complement to Godey & Artzi (2026)'s single-loss, norm-space account of the output head as a gradient bottleneck.

Roadmap. Section 4.1 establishes the diagnostic failure: on the rows that carry supervision the two gradients agree, while the aggregate the standard instrument reports does not. Sections 4.2 and 4.3 give the mechanism, an exact two-factor decomposition showing that a high-norm opposed minority, not disjoint support, sets the aggregate. Sections 4.5 through 4.7 give the account of the loss increase itself: the shared head makes the objective a cross-entropy against a blended target, so the optimum moves, which is why gradient surgery leaves the degradation untouched. Two further subsections carry load this summary would otherwise hide. Section 4.4 shows the signature is not specific to this loss pair. Section 4.8 withdraws a rare-token reading we can show is a construction artifact, and reports the 350M scale check.

2 Related Work

Multi-token prediction. Predicting several future tokens jointly dates to blockwise-parallel decoding (Stern et al., 2018) and ProphetNet (Qi et al., 2020). It was revived as a pretraining objective by Gloeckle et al. (2024), who attach n *independent output heads* to a shared trunk and report improved sample efficiency and up to 3× faster inference via self-speculative decoding. Two properties of their result are central to ours. First, each horizon gets its own *hidden state* and therefore its own logit vector, but the vocabulary projection is *shared*: their architecture is a shared trunk, n independent output heads implemented as transformer layers, and a single shared unembedding matrix. Second, the benefit is *more useful as model size grows*. DeepSeek-V3 (DeepSeek-AI, 2024) adopts sequential MTP modules at frontier scale and likewise shares the embedding and output head across depths, giving each depth its own transformer block and projection. Medusa (Cai et al., 2024) is the exception in this list: each of its heads carries its own vocabulary projection. Such extra heads are added primarily for speculative decoding (Samragh et al., 2025). MuToR (Gerontopoulos et al., 2025) interleaves learnable register tokens so future offsets can be predicted with no architectural change and negligible additional parameters. Across this line, the auxiliary signal is given *its own capacity*: separate heads, modules, or register tokens. Note what this does and does not separate. In the shared-unembedding designs above, gradients from every horizon still land on the same rows of one projection matrix; what per-horizon heads remove is the requirement that a *single* predictive distribution serve all horizons at once. Head-sharing therefore bundles two ingredients that can be separated: shared rows and a shared set of logits. The present work studies the minimal case in which both hold, a single set of logits produced by one tied projection serving every horizon. A separate-head variant would isolate the shared-label ingredient, since shared rows are present there too, and so would localize whether the effect is intrinsic to head-sharing (we identify this as the primary future experiment).

Small-scale degradation of MTP. Prior work documents small-scale degradation on *downstream benchmarks*: Gloeckle et al. (2024) find multi-token prediction increasingly useful at larger sizes and report their gains on HumanEval and MBPP, while Zuhri et al. (2025) report inconsistent improvements and underperformance on standard NLP benchmarks, and propose token-order prediction as an alternative. Whether that degradation also appears in next-token validation loss itself has not, to our knowledge, been reported at this scale. We measure it directly and find +0.39 nats. Both papers' smallest models (300M and 340M) also sit above the 124M at which we measure the gap, so this is a distinct observation rather than a restatement of theirs. Our contribution is not to add to the catalogue of when MTP helps or hurts. It is to measure *what the gradient geometry actually looks like* at the shared head when it hurts, and to show that the standard conflict diagnostic mischaracterizes it.

Gradient conflict and surgery. A large multi-task literature treats negative inter-task gradient cosine as the signature of interference and projects it away: PCGrad (Yu et al., 2020), Gradient Vaccine (Wang et al., 2021), and gradient-similarity loss weighting (Du et al., 2018), with later refinements (Sun et al., 2023; Shi et al., 2023). Related optimizers reformulate the conflict rather than projecting it directly: CAGrad (Liu et al., 2021) minimizes a worst-case objective within a neighborhood of the average gradient, and Nash-MTL (Navon et al., 2022) treats the update as a bargaining game among tasks. These methods operate on *global* (flattened) gradient vectors and target gradient *direction*. That the granularity of the measurement matters has been noted: Liu et al. (2025) report that model-level conflict resolution is poorly suited to their task and detect conflicts at a finer-grained modular level instead. Their remedy is finer-grained surgery. Our result is that at the output head finer-grained surgery is inert, so the granularity fix does not carry over here. A parallel line instead balances gradient *magnitude*: GradNorm (Chen et al., 2018) dynamically tunes per-task gradient norms rather than reprojecting directions. This magnitude-versus-direction distinction is central to our findings, which locate the operative structure in per-row gradient norms rather than in the direction the aggregate cosine reports. The Jiang et al. (2023) result noted in Section 1 points the same way from the auxiliary-task side. This connects to a broader empirical result: Kurin et al. (2022) and Xin et al. (2022) find that plain loss-summing (unitary scalarization) matches or beats specialized multi-task optimizers, which appear to act largely as regularizers rather than by resolving genuine conflict. Our surgery-null result is a mechanistic instance of their empirical claim: the opposition a conflict diagnostic detects at the output head is mass-dominant (the opposed rows carry 60–69% of the gradient-magnitude mass) yet causally inert, so a projection that removes exactly that opposition does not repair the degradation (Appendix A). The aggregate is set by per-row norm structure, and conflict surgery, though it has substantial mass to reproject, does not recover the baseline.

The output head as a bottleneck. Yang et al. (2018) showed the softmax output layer is a rank bottleneck on the distributions a model can express, and a line on *representation degeneration* and anisotropy (Gao et al., 2019) shows the learned output embeddings collapse toward a narrow cone, sharing direction across tokens. That shared-direction structure is one reason per-row gradients on the head look aligned, and it predicts the +1 median we observe on inactive rows. Godey & Artzi (2026) recently showed, for a *single* loss, that backpropagating through the low-rank output head destroys most of the gradient's norm, a norm-space single-loss bottleneck. Their object is the gradient flowing backward into the hidden state h through a rank-d projection; ours is the gradient deposited *on* the weight matrix W and its per-row structure, so the two results concern different tensors. Our result is the direction-space, *multi-loss* complement: we ask what a cosine diagnostic reports when two losses share that head, and find their per-row directions align while a high-norm opposed minority sets the aggregate (Section 4.3). The same group has since proposed separating a transformer's state-keeping and prediction roles into two computation streams, reporting a difference in the gradients the design entails (Monea et al., 2026). That is a design response; ours is a measurement of what a conflict diagnostic reports at the shared head. The two are complementary rather than competing.

Positioning. Relative to prior work, three points of contrast matter most; the paper's full contributions are the six listed in Section 1.

(i) We provide a per-row gradient measurement at the tied output projection that a global cosine misses.

(ii) We show that the next-token degradation of small-scale shared-head multi-token training, whose downstream counterpart is documented but unexplained (Gloeckle et al., 2024; Zuhri et al., 2025), co-occurs with *aligned* per-row gradients rather than with broad directional conflict. The aggregate is set by a high-norm opposed minority.

(iii) Our standard capacity-separated auxiliary *reduces* the next-token cost, in the same spirit as capacity-added fixes, though in our runs it does not fully recover the CE-only baseline within noise. This holds demonstrably for the standard variant; the tuned variant sits below our detection floor (Section 4.7) and is inconclusive.

Our contribution is therefore not a fix but the *diagnostic mechanism*: why conflict-based remedies act on a high-magnitude but causally inert opposition here, and why the auxiliaries that help are the ones that avoid the shared head.

3 Method

3.1 Models, data, and training recipe

We train GPT-2 (Radford et al., 2019) (124M; 12 layers, d=768, vocab 50,257 padded to 50,304) and a 350M-class variant (355M parameters; 24 layers, d=1024) on FineWeb-Edu (Penedo et al., 2024), tokenized with the GPT-2 BPE. The output projection (`lm_head`) is *tied* to the input embedding by default; Section 4.4 reports an untied control. The primary objective is next-token cross-entropy (CE). The auxiliary multi-token objective (MTP) supervises the *same tied head* to also predict tokens at offsets t+2 and t+3, with total loss $\mathrm{CE} + 0.5\,\mathrm{MTP}_2 + 0.25\,\mathrm{MTP}_3$. Evaluation is **pure next-token CE on held-out validation** for every variant, so all reported losses are directly comparable. The loss-comparison runs use a batch of 16,384 tokens/step for 30,517 steps (499,990,528 tokens processed, $\approx$ 4.0 tokens/parameter at 124M). The training slice is 480M unique tokens, so this is 1.04 epochs; the tokens-per-parameter figure counts tokens processed (3.86 if counted over unique tokens instead). This is an intentionally undertrained, small-batch regime, chosen for controlled comparison and because it is where shared-head multi-token training is already documented to underperform; we return to this scope in Section 6.

Unless stated otherwise every run uses the same optimizer recipe: a Muon/Adam hybrid (Jordan et al., 2024) that places the 2D hidden weights inside transformer blocks (Q, K, V, attention-out, and both MLP matrices; 84.9M of 124.4M parameters) under Muon at a flat lr=0.02, momentum 0.95, weight decay 0.01, and all remaining parameters (token and position embeddings, and the tied head) under Adam at lr=3×10^{-4}, β=$(0.9, 0.95)$, weight decay 0.1. Only the Adam group follows a schedule: 200-step linear warmup then cosine decay to 10% of base; the Muon group is held flat for the whole run. Training uses bfloat16 autocast. The token pool is the first 500M tokens of the streamed `sample-10BT` split, of which the final 20M are held

condition	n	mean	sd
CE-only (A)	5	3.9851	0.0138
G_{nce}	5	4.0241	0.0207
G_{nce} tuned†	3	4.0026	0.0156
NextLat	3	4.0365	0.0179
shared MTP (B)	3	4.3767	0.0209
MTP stop-gradient	3	4.4927	0.0101

Table 1: No auxiliary demonstrably recovers the CE-only baseline. Final validation loss by condition at 124M, Muon hybrid recipe, 30,517 steps. Standard deviations are across-seed sample statistics (ddof=1), never standard errors. Deltas quoted in the text are computed from unrounded values, so they can differ from differences of these rounded means by ± 0.001. †From the ablation driver, at G_{nce} layers $(10, 9)$ and multiplier m=0.1; the other rows are from the phase-B driver, where the one condition that has these hyperparameters at all (G_{nce}) uses layers $(9, 8)$ and m=0.3. As the preceding paragraph notes, neither of those two differences is resolvable at these seed counts.

out for validation; periodic evaluation averages 20 batches and the reported `final_val` averages 50, all at batch 16×1024. Appendix A reports one plain-AdamW replication arm (n=2 seeds across three conditions) that deliberately departs from this recipe, and its absolute losses are not comparable with the rest.

Auxiliary specifications. G_{nce} is an InfoNCE objective on intermediate hidden states: t+2 reads block 9 and t+3 reads block 8 (each through its own LayerNorm), targets are *detached* token embeddings, K=16 in-batch negatives drawn by rolling the batch, temperature τ=0.1, horizon weights 0.5 and 0.25 matching shared MTP, and a single auxiliary weight α calibrated once at the first step to $\min(m \cdot \text{CE}/\text{NCE}, 10)$ and floored at 0.01, with multiplier m=0.3. No term touches the output head.

The *tuned* variant comes from a separate ablation driver, and two settings differ from the standard condition, not one: it reads blocks 10 and 9 rather than 9 and 8, *and* it uses m=0.1 rather than 0.3. Neither term is resolvable at these seed counts, and we decline to attribute the difference to either. Two comparisons bound each term. Within the ablation family, matched seeds exist at both layer pairs under m=0.1. Switching $(9, 8) \rightarrow (10, 9)$ there moves the loss by -0.004: paired t=-0.40, p=0.73, n=3, with a 95% CI of $[-0.048, +0.040]$. Across families, lowering the multiplier at fixed layers $(9, 8)$ shifts it by 0.017, from 4.0241 at n=5 down to 4.0067 at n=3: Welch t=1.43, p=0.21, with a 95% CI of $[-0.013, +0.048]$. Both intervals contain zero. Both are also wider than the 0.021 tuned-versus-standard difference they are meant to decompose. That difference is therefore itself within run-to-run noise, consistent with the $p \approx 0.19$ reported against the CE-only baseline in Section 4.7. The two conditions also come from different run families with different seed counts, so the contrast is not a clean single-factor comparison in the first place. The configuration was selected by a 14-run one-factor-at-a-time screen at 5,000 steps over $m \in \{0.1, 0.3, 1, 3\}$, layer pairs $\{(10, 9), (9, 8), (8, 7)\}$, $\tau \in \{0.05, 0.1, 0.2, 0.5\}$, $K \in \{1, 4, 16\}$, roll-versus-random negatives, and horizon weights $\{(0.5, 0.25), (0.25, 0.125), (1, 0.5)\}$, then rerun to 30,517 steps at n=3. The screen scored on the same held-out split used for the reported losses, so the tuned condition's 4.003 is optimistically biased and its Δ+0.018 inherits that bias; we treat that comparison as an upper bound on how close a tuned auxiliary comes, not as evidence of recovery. NextLat is due to Teoh et al. (2025), who introduce next-latent prediction as a self-supervised auxiliary in latent space and report that it leaves the architecture, training efficiency and inference of the base transformer unchanged. Our arm is a reimplementation at this study's 124M scale, written to test a head-free auxiliary rather than to reproduce theirs; we have not verified which implementation details agree, so no quantitative comparison to their results is drawn here. The specification we ran is the one given next. NextLat predicts the *next-position latent*: a bias-free linear map on the LayerNorm'd block-11 output at position t against the detached block-11 output at t+1, with loss $1 - \cos$ and the same α calibration. MTP stop-gradient keeps the 0.5 and 0.25 weights and differs from shared MTP only in forming the auxiliary logits as $\text{sg}(W)h$.

Per-condition losses. Table 1 gives the mean, sample standard deviation and n for every 124M condition, so the Welch comparisons in Section 4.7 can be reproduced from the table alone.

Run labels used below follow the scripts in the released repository: A denotes the CE-only condition and B the shared-MTP condition (so "B@42" is shared MTP at seed 42), A1–A4 are the phase-A diagnostic measurements (A1 Muon, A2 AdamW, A3 the control, A4 the untied-head control), and phase-C denotes the negative-control and surgery runs of Appendix A.

Three uses of "AdamW". The name appears in three distinct roles and we keep them separate throughout. (i) Inside the Muon hybrid, an Adam group with weight decay carries the embeddings, positional table and output head. (ii) The AdamW *diagnostic* runs (A2 and the three AdamW norm-support seeds) place every parameter under a single `torch.optim.AdamW` with lr=10^{-3}, β=$(0.9, 0.95)$, ϵ=10^{-8}, weight decay 0.01 on the decayed group and 0 on norms and biases, tied weights deduplicated, and all groups scheduled; these produce the -0.28 aggregate. (iii) The plain-AdamW phase-C training recipe of Appendix A uses lr=3×10^{-4} and no weight decay; its absolute losses are not comparable with the rest, so the -0.28 diagnostic aggregate should not be mapped onto that replication.

Seed counts differ by condition and are fixed by what was run: CE-only and G_{nce} use n=5 (seeds $42, 123, 456, 789, 1337$); shared MTP, MTP stop-gradient, NextLat and tuned G_{nce} use n=3 (seeds $42, 123, 456$). Paired comparisons therefore use the three seeds common to both arms; Welch comparisons use all available runs per condition.

3.2 Measuring per-row gradients at the output projection

The output projection has shape (V, d); row i is the weight vector for vocabulary token i. For a validation batch we compute the gradient each loss deposits on this matrix and, for **each row**, the cosine between the CE and MTP row-gradients. We summarize the resulting distribution of V per-row cosines by three statistics: its median, the fraction with $\cos < 0$ (directional opposition), and the fraction with $|\cos| > 0.3$ (appreciable *association* rather than alignment). The absolute value counts an anti-aligned row alongside an aligned one. We therefore use the signed $\cos > 0.3$ and $\cos < 0$ wherever the direction of the relationship is what matters, and reserve $|\cos|$ for distance-from-zero statements. The 0.3 threshold is a conventional "appreciable correlation" cut. The qualitative picture is insensitive to it, since the active-row cosines cluster around 0.5, well above 0.3, while the parallel majority sits at $+1$. Each gradient measurement uses a single held-out batch of 16 sequences $\times$ 1024 tokens (16,384 tokens); Section 4.1 reports how the active-row fraction varies with batch count. We contrast these per-row statistics with the **aggregate** cosine (the cosine between the two *flattened* matrix gradients), which is the quantity conventional conflict diagnostics use. The 47 padding rows (50,304 − 50,257) are masked from all per-row statistics.

Measurement windows (stated explicitly). The gradient instrumentation and the loss comparison come from *different* runs, and we are careful to keep them distinct. At 124M, the per-row/aggregate trajectory is a dedicated diagnostic run logged densely over its first 1000 steps (A1). The diagnostic runs A1, A2 and A4 are themselves trained on the shared-MTP objective (`loss_mode=mtp_shared`), so the per-row cosines are measured inside a model that is being trained with the auxiliary, not on a CE-only model with the MTP gradient merely evaluated; the one exception is the CE-only-trained comparator at 350M in Section 4.8, and a matching CE-only 124M diagnostic is the cheapest of our future items. The +0.39-nat loss gap is the separate 30,517-step training comparison (Section 4.7). Late-training persistence of the alignment is therefore *not* established at 124M; it is measured directly only at 350M (single seed), where per-row $|\cos| > 0.3$ remains high out to step 87,000 (Section 4.8). We flag throughout which quantity comes from which run, and treat single-seed/single-run measurements as such.

Sensitivity to the ratio tolerance. The active-row classification uses $|\|g_i^{\text{mtp}}\|/\|g_i^{\text{ce}}\| - 0.75| > 0.01$. Over the three Muon seeds, widening or tightening that tolerance moves the active fraction and the active-row median only slowly: at tolerance 0.005, 0.01 and 0.02 the active fraction is 8.5%, 8.1% and 7.6% and the median cosine 0.536, 0.529 and 0.525. Only the far tighter 0.001 behaves differently (14.2% active, median 0.785), because at that level the classifier admits rows whose norm ratio deviates only slightly from 0.75. We cannot tell from this measurement whether those deviations are float noise or the embedding-path contribution discussed below; the instrument does not separate them. The reported conclusions do not turn on the exact tolerance.

Which gradient the instrument reads. The measurement calls `torch.autograd.grad` on `lm_head.weight`, which in this model *is* the token-embedding tensor (`model/gpt2.py` ties them). The returned gradient therefore sums two paths: the output-projection path $\sum_t \delta_i(t)h_t$, and the embedding path, which deposits on row i at every position where token i is an *input*. The 0.75 identity of Section 4.1 is derived for the projection path alone.

The two paths land on almost exactly the same rows, and that fact cuts both ways. Next-token cross-entropy scores position $t+1$ from input t, so a token that is an input anywhere in the batch is a target one position earlier, with one exception per sequence (the first token). Input rows are therefore a subset of target rows up to that exception. The reassuring consequence is that our active/parallel partition does not meaningfully conflate the two populations: the input set adds essentially nothing to the target set, and we do not claim otherwise. The unwelcome consequence is that the embedding path deposits *only* on rows that are already active, which is precisely where every statistic in this paper lives. Its contribution to the active-row cosines is not bounded by anything we measure, and a check on the parallel rows could not bound it even in principle, because those rows receive no embedding-path gradient at all. We state this as an open limitation rather than a quantity we have controlled. The cheapest way to close it is to recompute the projection path alone, or to commit per-row norms from an untied run, where the head tensor receives the projection path only; we list both in Section 6.

Gradients, not updates. Both our instrument and the conflict-surgery literature operate on raw loss gradients, obtained here by `torch.autograd.grad` before any optimizer transform. The optimizers do not apply those vectors directly: Muon orthogonalizes the update for 2D hidden weights and Adam preconditions elementwise. Note that in both recipes the head itself sits under Adam(W): the hybrid assigns only the hidden 2D weights to Muon. The optimizer-dependence we report therefore enters through the trunk-shaped hidden states and the differing training trajectories, not through the head's own preconditioner. That said, our magnitudes are consistent with the gradient-versus-update gap mattering, and a diagnostic defined on post-transform updates could report differently; we measure gradients because that is what the surgery methods act on.

Terms used throughout. Table 2 collects the paper-specific quantities. Each is defined where it is first used; the table is a reference, not a substitute for those definitions.

3.3 Separating directional conflict from disjoint support

The distinction the paper turns on: the **aggregate** cosine measures whether the summed gradient vectors point the same way; the **per-row** distribution measures whether they agree row by row. These can disagree, and the reason is that the aggregate cosine is *magnitude-weighted* (it is the cosine of the summed, flattened gradients) whereas the per-row median and the cos < 0 fraction are *unweighted* over rows. A near-zero or negative aggregate with a per-row median of $+1$ is therefore consistent with (a) *support divergence*, where the two losses place their magnitude on disjoint rows, so the aggregate drifts toward orthogonality without any directional opposition; or with (b) a small number of *high-norm opposed rows* that dominate the weighted sum while the many low-norm rows align. The 0.33% opposed-by-count figure (measured in Section 4.1) does not bound the opposed *mass*, so counting rows cannot distinguish the two. Separating them requires the per-row gradient *norms*; we measure them directly (Section 4.3, Figure 4) and find (b).

A four-row illustration. Take a weight matrix with four rows. Let three low-norm rows be aligned, $g_i^{\mathrm{ce}} = g_i^{\mathrm{mtp}} = (\epsilon, 0)$ for $i = 1, 2, 3$ with small ϵ, and let one high-norm row be *opposed*, $g_4^{\mathrm{ce}} = (1, 0)$ and $g_4^{\mathrm{mtp}} = (-1, 0)$. Every per-row cosine is defined: three read $+1$ and one reads -1, so the median and the cos < 0 fraction (1/4) both say "mostly aligned." Yet the flattened vectors are dominated by row 4, so the **aggregate cosine is negative** (≈ -1 as $\epsilon \to 0$). No support divergence is needed: the losses occupy the *same* rows, but one high-norm row is opposed and it sets the sign. This is mechanism (b), and it is what we measure at scale.

term	meaning
active row	A vocabulary row classified as receiving a target in the measurement batch, by the norm-ratio proxy $\lvert \lVert g_i^{\text{mtp}} \rVert / \lVert g_i^{\text{ce}} \rVert - 0.75 \rvert > 0.01$. A row that receives a target moves the ratio away from 0.75; the converse does not hold exactly (Section 4.1).
parallel row	A row failing that test. Absent a target, the two gradients are scalar multiples with per-row cosine +1 by construction, which is why they must be separated before any per-row statistic is read.
f, parallel-mass fraction	The share of total gradient-magnitude mass carried by parallel rows. Small ($\approx 3\%$), which is why the full-vocabulary and active-row denominators give nearly the same mass-weighted answers.
full-vocabulary vs. active-row denominator	Whether a statistic averages over all V rows or only active ones. Count-based statistics (median cosine, opposed fraction) change materially between the two; mass-weighted ones barely move. Every reported statistic names its denominator.
aggregate cosine	The cosine between the two losses' summed, flattened gradients over the whole matrix: the quantity a standard conflict diagnostic reports. Magnitude-weighted (Section 3.3).
mass-weighted cosine, $\langle \cos \rangle_{\text{mass}}$	The per-row cosines averaged with weights $w_i \propto \lVert g_i^{\text{ce}} \rVert \, \lVert g_i^{\text{mtp}} \rVert$. One of the two exact factors of the aggregate in Eq. (1).
norm-profile cosine, ρ_{norm}	The cosine between the two losses' vectors of per-row gradient *norms*. Near 1 means both losses place their magnitude on the same rows; near 0 would mean disjoint support. The other exact factor of the aggregate.
opposed-norm fraction	The share of gradient-magnitude mass carried by rows with negative per-row cosine. Distinguishes "few opposed rows" from "opposed rows that matter."
anchor gate	The check, fixed in advance, that the sweep's s=0 and s=1 endpoints reproduce the CE-only and shared-MTP conditions within $\pm$0.04 nats (Section 3.5).
coverage	The probability mass retained by top-k truncation when estimating the KL. Lower coverage biases the estimate upward, which we handle by extrapolating linearly in the uncovered mass.
band	The span of the KL prediction over {offset-3 treatment} $\times$ {validation half} at the largest k, reported instead of a point estimate because the two offset-3 treatments bias in opposite directions.

Table 2: Paper-specific terms. Entries carry a section reference where the term is defined outside this table; the rest are defined here. The active-row rule and the two exact factors of the aggregate carry most of the argument's weight.

3.4 Calibrating the instrument against an unrelated-loss control

To confirm the instrument is not trivially reporting alignment, we run a control (A3) that measures CE against an unrelated L1 regularizer on the same head. A well-calibrated instrument should read near-zero alignment for two unrelated losses.

In $x \pm y$, x is the mean over seeds and y is the sample standard deviation across them (ddof=1), throughout, including in the statistical tests. Spreads are never standard errors.

3.5 Auxiliary-weight sweep and the KL estimator

Two measurements in Section 4.6 need their protocol fixed in advance. First, the **weight sweep**: we retrain from scratch at auxiliary scales $s \in \{0, 0.25, 0.5, 1.0\}$ with loss CE + $s\,(\frac{1}{2}\text{MTP}_2 + \frac{1}{4}\text{MTP}_3)$, seed 42, every other hyperparameter held at the values above, so that s=0 and s=1 reproduce the CE-only and shared-MTP conditions and act as anchors. We fixed the anchor tolerance in advance at $\pm$0.04 nats (roughly 2 shared-MTP within-condition standard deviations) before running the sweep. The comparator is the matched seed-42 run at each end, not the condition mean, since the sweep is a single seed. Both endpoints pass on that reading, the s=1 end by the narrower margin: 0.034 against the 0.04 allowance, or 84% of it.

Second, the **KL estimator**. The soft-label identity of Section 4.5 predicts a next-token penalty $\mathrm{KL}(P_1 \| q_s^*)$ that depends only on the data distribution, so we estimate it from the s=0 checkpoint alone and compare it to the sweep without fitting anything. $\hat{P}_1$ is the model's own next-token distribution; $\hat{P}_2$ is obtained by rolling the model forward one step over its top-k next-token candidates and marginalizing, renormalized over the retained mass, with $k \in \{256, 1024, 2048\}$. Truncation biases the estimate upward relative to full marginalization, which we handle by reporting the coverage attained and extrapolating linearly in the uncovered mass. Offset 3 cannot be reached this way at acceptable cost, so we *bracket* it: one variant drops the t+3 term (under-weighting future mass, biasing the prediction down) and one sets $P_3 \approx P_2$ (substituting a too-peaked component, biasing it up). We split the held-out set in half to expose estimator variance, and report as our band the span over {offset-3 treatment} × {validation half} at the largest k, rather than a point estimate. Smaller k gives systematically higher estimates (truncation is an upward bias), so pooling across k would widen the band with values we already know to be biased; we quote the smaller-k estimates separately where the coverage trend matters. The one remaining known bias, proxy smoothing from an undertrained model, is downward and we do not have a way to quantify it; we say so where it matters rather than correcting for it.

4 Experiments and Results

4.1 Per-row gradients are predominantly aligned on the rows that carry supervision

On the vocabulary rows that receive a target in the measurement batch, the next-token and auxiliary gradients point in nearly the same direction. That is the finding of this subsection, and it is the opposite of what the aggregate cosine reports. Table 4 collects the per-row and per-norm statistics this subsection and Section 4.3 establish, so the numerical claims below can be checked against a single place.

The honest denominator: active rows. Because CE and the shared MTP loss read the *same* logits from the *same* hidden state, a vocabulary row that is not a target anywhere in the measurement batch has $g_i^{\mathrm{mtp}} = 0.75\, g_i^{\mathrm{ce}}$ up to sequence-boundary terms. The two objectives differ only in their target-indicator terms, and $0.75 = 0.5{+}0.25$ is the summed auxiliary weight. The identity is exact on the common position set for the output-projection path; because the head is weight-tied, the measured gradient also carries an embedding-path term, which Section 3.2 bounds only for rows that carry no target. The last one or two positions of each sequence carry no t+2/t+3 target, which we do not mask, so the correction is ≤2 positions per 1,024. The empirically measured norm ratio $\|g_i^{\mathrm{mtp}}\|/\|g_i^{\mathrm{ce}}\|$ on such rows is 0.750, which is itself a check that all horizons use the same hidden state. Such rows are scalar multiples with cosine identically +1, and they dominate the count. We therefore classify rows by a proxy for *target presence*: a row is *active* iff $\big|\|g_i^{\mathrm{mtp}}\|/\|g_i^{\mathrm{ce}}\| - 0.75\big| > 0.01$. A row that receives a target moves the ratio away from 0.75. The converse does not hold exactly: a row whose ratio coincidentally sits within the tolerance is counted parallel regardless. We quantify the size of that approximation below. The test is on the norm ratio, not on the cosine, so it does not presuppose the alignment it is used to measure.

By this classification ≈92% of rows are parallel and only ≈8% **are active** (the exact fraction is batch-relative; see the batch-count check below). Restricted to the active rows, the median per-row cosine is ≈0.53 (Muon 0.529±0.008, AdamW 0.547±0.016, n=3), ≈3–4% are opposed, and ≈83% have cos > 0.3, against 1.0% for the CE-vs-L1 control on the same denominator. Over the full vocabulary the same run reads a median of +1.00, 0.33% opposed (the phase-A diagnostic batch; the per-row-norm batch of Figure 5 gives 140 rows, 0.28%, on the same seed and step), and 98.4% above $|\cos| > 0.3$ (Figure 2, also phase-A); these are the construction-inflated numbers, reported only to show what the full-vocabulary denominator does to the same measurement. Table 3 reconciles the two denominators run by run rather than asserting that they agree.

The active-row median of ≈0.53 is the honest measure of directional alignment, and it is still well above zero, in sharp contrast to the near-zero aggregate. Crucially, the active-row correction does *not* affect the mechanism in Section 4.3, which is mass-weighted. The parallel rows carry only a few percent of the gradient-magnitude mass (3.1% averaged over the five conforming runs below, 4.5% including the outlying seed). The norm-profile cosine (0.98) and opposed-norm fraction (0.60/0.69) are therefore essentially unchanged whether

optimizer	seed	F (%)	a (%)	reconstructed (%)	measured (%)	parallel rows below 0.3
Muon	123	98.57	8.18	82.5	82.4	24
Muon	42	98.66	7.82	82.9	83.1	43
Muon	456	97.72	8.36	72.7	81.3	404
AdamW	123	98.63	8.02	82.9	83.0	36
AdamW	42	98.82	7.70	84.6	84.9	41
AdamW	456	98.48	8.21	81.5	81.5	35

Table 3: **The two denominators reconcile, but only approximately.** F is the full-vocabulary fraction with $|\cos| > 0.3$; a is the active fraction. Treating every parallel row as clearing the threshold gives the reconstruction $(F-(100-a))/a$, compared against the directly measured active-row fraction with $\cos > 0.3$. The two thresholds differ in sign convention. The gap between them is the active-row population with $\cos < -0.3$, measured at 0.75–0.96% of active rows across the six runs. That is not negligible at the precision of this check, so part of the residual below is the convention difference rather than reconstruction error. Five runs agree within 0.3 points on the signed comparison (1.1 points against the unsigned active fraction). Percentages are computed from unrounded row counts. F and a are displayed to two decimals and the reconstruction and measured columns to one, so recomputing from the displayed operands can differ in the last digit. Muon seed 456 falls short by 8.7 because 404 of its parallel rows sit below $|\cos| = 0.3$ and are charged to the active set by the subtraction, against 24–43 in the other five runs. The reconstruction is therefore a consistency check, not an identity.

computed over all rows or active rows only. Restated on the active-row denominator the norm-profile cosine is 0.978 (Muon) and 0.976 (AdamW), against 0.979 and 0.977 full-vocabulary, and the opposed-norm fraction is 0.608 ± 0.020 and 0.708 ± 0.010, against 0.604 ± 0.026 and 0.692 ± 0.011 (sample sd, ddof=1, as everywhere; the same four quantities appear per seed in Appendix A). An opposed row must carry a target: rows that are parallel by construction have $\cos = +1$ exactly. So under the *true* target-presence partition, restricting the denominator removes only parallel mass and the relation active $=$ full$/(1-f)$ holds exactly, with f the parallel-mass fraction. The ratio classifier only approximates that partition, and departures from the relation measure exactly the opposed mass it misfiles, which makes the identity a diagnostic for the classifier rather than a claim about the data. Measured per seed, f is 2.3–3.9% (Muon) and 2.8–3.5% (AdamW) in five of the six runs, where the relation reproduces the active-row value to within 0.008; the sixth is the exception below. Muon seed 123 shows $f = 11.5\%$ and the only negative shift (-0.025), because five rows land inside the 0.01 ratio tolerance despite being opposed: the classifier tests magnitude only, and on these rows the magnitude ratio happens to sit at 0.75 while the two directions differ. One of those five rows is the newline token, carrying 8.7% of all mass at $\cos = -0.86$ on that seed (Figure 5 shows the same token on seed 42, at 10.8% and $\cos = -0.91$). The classifier is therefore a generic rather than exact proxy for target presence: a row whose norm ratio coincidentally sits near 0.75 is counted parallel. At tolerance 0.005 that seed's misclassified mass falls to zero and its shift turns positive ($+0.009$), which is the direction the relation requires. The -0.025 shift is therefore the diagnostic working as intended, not a violation of it. None of this moves the mechanism: the shifts are at most a few percentage points of a fraction whose distance from 0.5 is what the claim rests on. The interference lives among the active rows; restricting to them makes the finding sharper, not weaker.

Accumulating measurement batches grows the active set as expected while leaving the active-row statistics stable. This check is a separate instrument from the seed means above, run on a single seed with a different batch draw, so its medians are not directly comparable to the $\pm$ spreads quoted for n=3: one batch gives 8.9% active rows with median cosine 0.541 and 3.0% opposed, and two batches give 14.5% active with median 0.539 and 3.1% opposed. The active *fraction* is therefore batch-relative, as it must be (more batches place targets on more rows), while the active-row median and opposed fraction are not artifacts of a single batch draw.

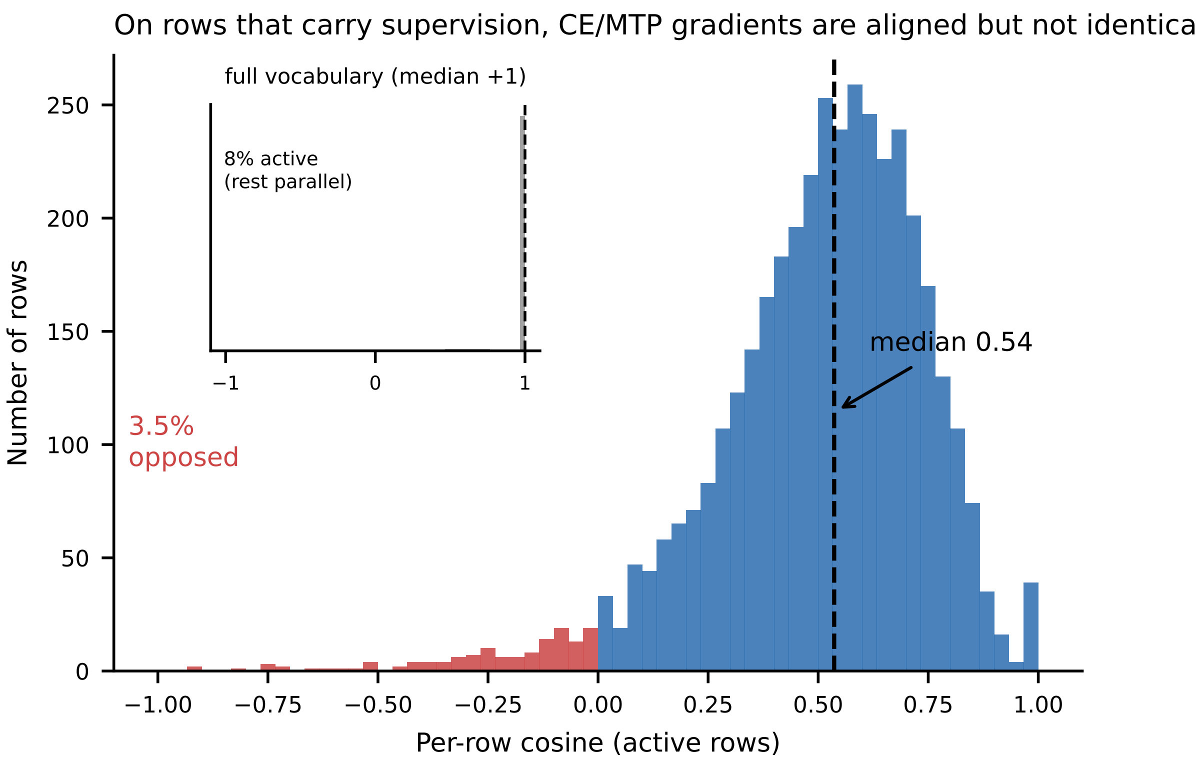


Figure 2: **Per-row CE/MTP gradients on the output projection are aligned on the rows that carry supervision.** Main panel: distribution of *signed* per-row cosines over the active rows (median ≈0.53; ≈3–4% opposed, in red). Inset: the full-vocabulary distribution, whose median is +1.00 because ≈92% of rows are parallel by construction ($g^{\mathrm{mtp}} = 0.75\, g^{\mathrm{ce}}$ for any row with no target in the batch, since CE and the shared MTP loss read the same logits). On the matched active-row denominator, 83% of active rows have $\cos > 0.3$ against 1.0% for a CE-vs-L1 control measured on the same rows. Full-vocabulary control readings vary across snapshots (0.4% on the original diagnostic run, 36% on the sweep snapshot, both measured with the released instrument; at the sweep snapshot the full-vocabulary control median is −0.24, so that reading is structured rather than merely noisy), which is why we do not rest calibration on the full-vocabulary denominator. Muon, seed 42, GPT-2 124M, step 1000, 50,257 vocab rows (47 padding rows masked); this single draw reads 7.8% active, median 0.536, 3.5% opposed, and the panel annotations are that draw rather than the seed-averaged values (0.529 ± 0.008, $3.7 \pm 0.2\%$ over three Muon seeds). The active fraction is 7.7–8.4% across all six norm-support runs; the separate batch-robustness check of Section 4.1, which accumulates over more batches, reads higher (8.9% at one batch, 14.5% at two). Accumulation explains the rise to 14.5%, since more rows receive a target as batches are added; the single-batch 8.9% sits above the seed range only because it is a different batch draw. The mass-weighted mechanism (Section 4.3) is unaffected by the active-row restriction because parallel rows carry only ≈4% of the gradient mass on this seed (3.89%; see Section 4.1 for the range across runs).

4.2 The aggregate cosine decouples from per-row alignment during training

The per-row picture and the aggregate one do not merely differ at the end of training; they start in agreement and come apart. Despite the aligned per-row median of Section 4.1, the aggregate cosine reads ≈ 0 (−0.08 at step 1000). Tracking both metrics over training (Figure 3) shows this is not a fixed property. At initialization the aggregate and per-row alignment agree at $\approx +0.99$. The aggregate then crosses zero early, between steps 100 and 150 under Muon and between 150 and 200 under AdamW, while the per-row median holds at $\approx +1$ for the remainder of the 1000-step diagnostic window. Late-training persistence is shown separately at 350M-class scale (Section 4.8). These trajectories are a single diagnostic run per optimizer; the close agreement between two different optimizers is the replication we have.

We can pin down *how* the aggregate collapses without any additional runs, using an exact identity. The aggregate cosine is a magnitude-weighted average of the per-row cosines, agg $= \sum_i w_i \cos_i$ with weights $w_i = \|g_i^{\text{ce}}\|\|g_i^{\text{mtp}}\|/(\|G^{\text{ce}}\|\|G^{\text{mtp}}\|) \geq 0$. If gradient magnitude were spread evenly across rows, the weights would be equal and the aggregate would equal the *plain mean* of the per-row cosines. That mean is $+0.94$ (Muon) and $+0.96$ (AdamW), whereas the actual aggregate is -0.08 and -0.28 (Figure 6). The entire ≈ 1.0–1.2 gap between "aggregate if norms were flat" and the real aggregate is therefore attributable to the per-row *norm* structure, not to the typical per-row direction. The standard diagnostic is reporting a magnitude-weighting artifact, which is the paper's central point made quantitative.

The weights are unnormalized, and their sum is itself informative. Writing $a_i = \|g_i^{\text{ce}}\|$ and $b_i = \|g_i^{\text{mtp}}\|$, we have $\sum_i w_i = \langle a, b\rangle/(\|a\|\|b\|) = \cos(a, b)$, which is exactly the *norm-profile cosine* ρ_{norm} defined below. The aggregate therefore factors exactly as

$$\text{agg} = \sum_i w_i \cos_i = \underbrace{\Big(\sum\nolimits_i w_i\Big)}_{\rho_{\text{norm}}} \cdot \underbrace{\frac{\sum_i w_i \cos_i}{\sum_i w_i}}_{\langle\cos\rangle_{\text{mass}}} = \rho_{\text{norm}}\,\langle\cos\rangle_{\text{mass}}, \tag{1}$$

the product of how much the two losses' norm profiles overlap and the mass-weighted mean of the per-row cosines. Evaluated within a single measurement, where all three quantities come from the same batch, this reads $-0.193 = 0.978 \times (-0.197)$ (Muon, seed 42) and $-0.319 = 0.985 \times (-0.324)$ (AdamW, seed 42). The identity is exact only within one measurement: the phase-A diagnostic aggregates (-0.08, -0.28) come from a different batch that did not retain per-row norms, so they cannot be factored directly. The two "disambiguating statistics" we introduce next are therefore not competing hypotheses but the two exact factors of the aggregate: support overlap is the total weight ρ_{norm}, and opposition is carried by $\langle\cos\rangle_{\text{mass}}$.

4.3 The aggregate is a norm-weighted cancellation, not disjoint support

The decomposition of Eq. (1) splits the aggregate cosine into two exact factors. The norm-profile cosine ρ_{norm} is the cosine between the two losses' vectors of per-row gradient *norms*, so it measures whether they load the same rows. The mass-weighted mean cosine $\langle\cos\rangle_{\text{mass}}$ measures whether the rows they load agree in direction (Table 2). What that identity does *not* settle is which of the two is responsible. A negative aggregate requires the opposed minority ($\cos < 0$, $\approx 0.33\%$ of rows) to carry *some* weight, so two mechanisms are a priori compatible with the aligned-median-plus-negative-aggregate pattern: (a) *support divergence*, where CE and MTP load their magnitude onto increasingly disjoint rows, making the products w_i small precisely where each gradient is large; and (b) a small number of *high-norm opposed rows* dominating the weighted sum. In terms of the identity above, (a) is $\rho_{\text{norm}} \to 0$ and (b) is $\rho_{\text{norm}} \approx 1$ with $\langle\cos\rangle_{\text{mass}} < 0$. Separating (a) from (b) needs the per-row gradient *norms*, which the earlier runs do not store. We measure them directly with a drop-in instrument (`measure_norm_support.py`) that logs per-row norms and reports two disambiguating statistics on the same short diagnostic pass. The *norm-profile cosine* is the cosine of the two per-row norm vectors: ≈ 0 under support divergence, ≈ 1 if the losses load the same rows. The *opposed-norm fraction* is the share of $\|g_i^{\text{ce}}\|\|g_i^{\text{mtp}}\|$ mass on rows with $\cos < 0$. Across three seeds per optimizer, the norm-profile cosine is 0.98. Because both vectors are nonnegative and both track token frequency, the scale needs a reference. Permuting one profile within ten contiguous token-id blocks, which retains coarse id-block structure but destroys the row-level pairing, gives 0.085–0.111 across the six runs. Global permutation gives 0.031. Both nulls depend on the permutation draw; we report single-draw values and the block range across runs, and neither is regenerable from the committed scripts, so they should be read as order-of-magnitude references rather than reproducible statistics. Each block still spans a wide frequency range, so this control does not hold the frequency profile fixed and the residual gap to 0.98 is not attributable to pairing alone. It does establish that nonnegativity and coarse frequency ordering together account for only $\approx$0.1 of the observed value, which is what the disjoint-support comparison needs. Per optimizer (Muon 0.979 ± 0.003, AdamW 0.977 ± 0.007): CE and MTP load the *same* rows, so support does *not* diverge. Yet the opposed-norm fraction is 0.60 ± 0.03 (Muon) and 0.69 ± 0.01 (AdamW): the majority of the magnitude mass sits on the

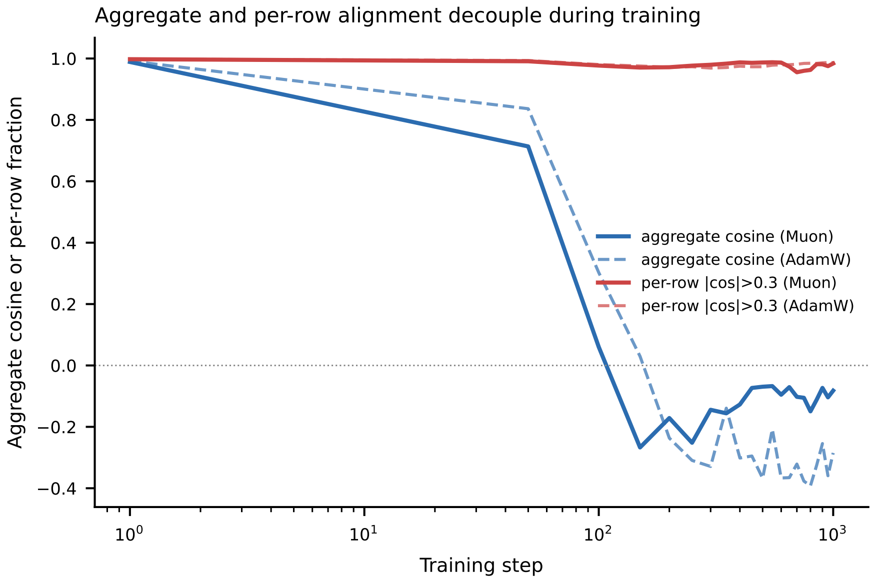


Figure 3: **The aggregate and per-row alignment start together and decouple.** Both metrics begin at $\approx +0.99$; within $\sim$200 steps the aggregate (matrix-level) cosine falls below zero (crossing between steps 100–150 for Muon and 150–200 for AdamW) while per-row alignment stays high. The per-row curve is the full-vocabulary $|\cos| > 0.3$ fraction (≈ 0.98); on active rows the signed $\cos > 0.3$ fraction is ≈ 0.83 (Section 4.1; the unsigned fraction is slightly higher), but the decoupling from the aggregate is identical: the aggregate goes negative under either denominator. One diagnostic run per optimizer (Muon solid, AdamW dashed); the agreement between two optimizers is the replication. Both series share one axis: the aggregate cosine is a cosine in $[-1, 1]$ (blue), while the per-row series is a *fraction* of rows in $[0, 1]$ (red), so the red curves near 0.98 are fractions and not cosines.

opposed minority. Mechanism (b) holds, and it holds under both optimizers, with AdamW canceling more strongly than Muon (Figure 4). The difference in opposed-norm fraction (0.69 vs 0.60) is significant across seeds (Welch t=5.4, $p \approx 0.015$), so the magnitude is genuinely optimizer-dependent while the mechanism is shared. The near-zero aggregate is a norm-weighted cancellation by a high-norm opposed minority, not disjoint support.

Which rows carry the cancellation. The opposed mass is not spread thinly: on the Muon seed-42 pass, 140 opposed rows (0.28% of the vocabulary) carry 60.7% of the total $\|g_i^{\mathrm{ce}}\|\|g_i^{\mathrm{mtp}}\|$ mass, and just 47 rows account for two-thirds of *all* gradient mass (Figure 5, left). Decoding those rows shows they are the most frequent function words and punctuation: the four highest-mass opposed rows are `.`, the newline, `the`, and `,`, which alone carry $\approx$41% of the mass, followed by `of`, `to`, `and`, and `a` (Figure 5, right). Deleting those four rows reverses the sign of the aggregate. They are the same four token ids in all six runs (`,`, `.`, the newline and `the`), so this is not a per-seed selection. Recomputing the mass-weighted mean per-row cosine without them moves it from [−0.20, −0.14] to [+0.18, +0.25] under Muon, and from [−0.35, −0.30] to [+0.02, +0.12] under AdamW. The aggregate of Eq. (1) itself, which is the quantity a practitioner reads, moves from [−0.19, −0.13] to [+0.17, +0.24] under Muon and from [−0.34, −0.30] to [+0.02, +0.11] under AdamW. Each interval is the spread across the three seeds. Four rows are 0.008% of the vocabulary. The sign of the standard diagnostic is therefore set by a handful of punctuation and function-word rows, not by a broad property of the two gradients. This is the expected locus: these tokens are ubiquitous next-token *and* future targets, and their next-token and future-token continuation distributions genuinely differ, so the CE and MTP gradients on their output rows point in different directions while both being large. The cancellation is concentrated on a handful of high-frequency rows, not diffuse across the tail.

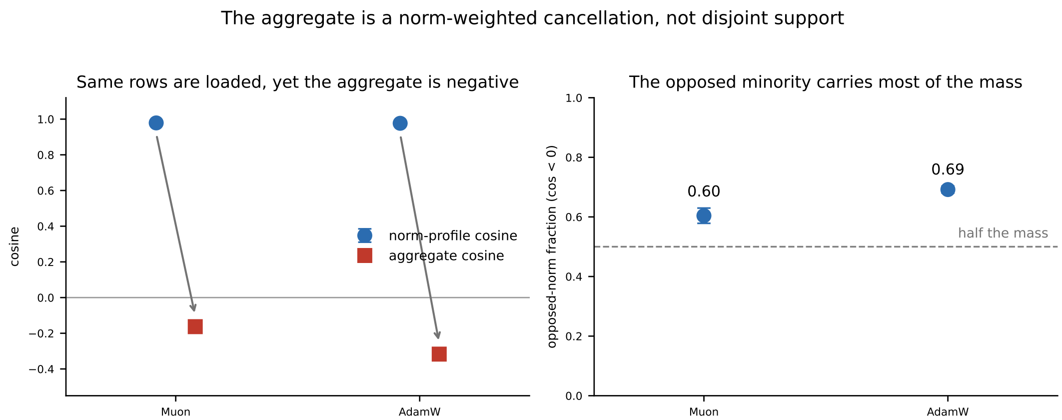
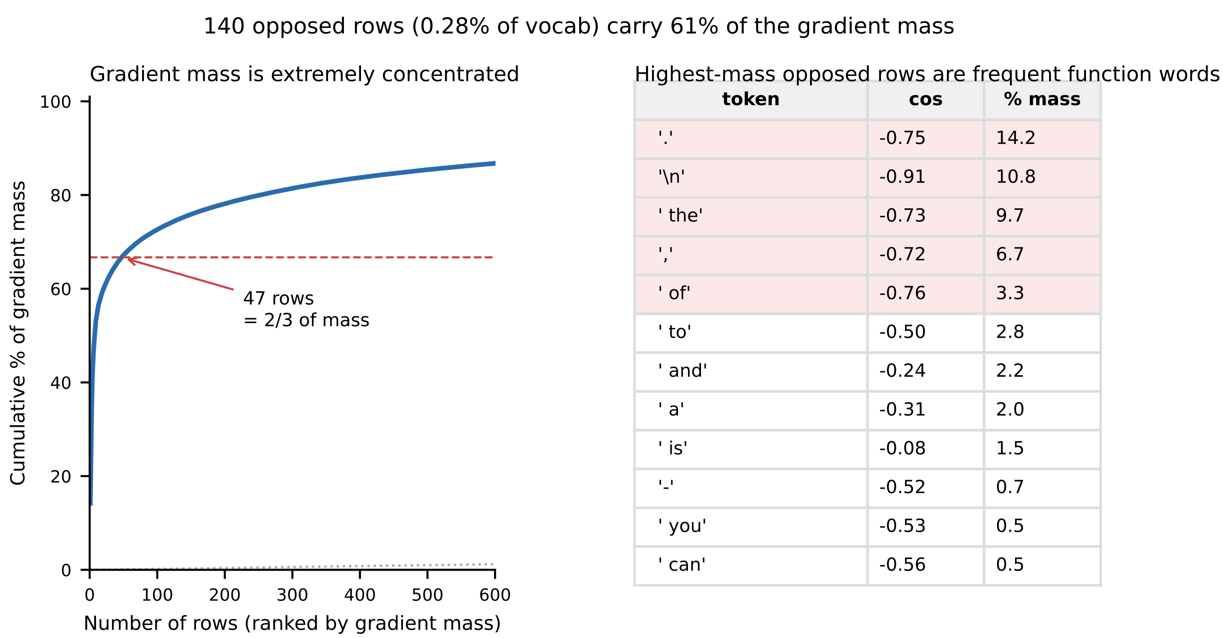


Figure 4: **The aggregate is set by a high-norm opposed minority, not by disjoint support** (n=3 seeds per optimizer, step 1000, 50,257 real vocabulary rows). *Left:* the norm-profile cosine (the cosine of the two per-row gradient-norm vectors) is ≈ 0.98 for both optimizers, so CE and MTP load the same rows; support does not diverge. The actual aggregate cosine nonetheless sits negative, and the arrow marks the collapse. *Right:* the opposed-norm fraction, the share of $\|g_i^{\mathrm{ce}}\|\|g_i^{\mathrm{mtp}}\|$ mass on rows with cos < 0, is above one half for both optimizers (0.60 Muon, 0.69 AdamW): the opposed minority carries most of the magnitude mass. Both panels use signed quantities, so neither conflates opposed rows with aligned ones. AdamW cancels more strongly than Muon; the mechanism is identical, the magnitude optimizer-dependent.



token	cos	% mass
'.'	-0.75	14.2
'\n'	-0.91	10.8
' the'	-0.73	9.7
','	-0.72	6.7
' of'	-0.76	3.3
' to'	-0.50	2.8
' and'	-0.24	2.2
' a'	-0.31	2.0
' is'	-0.08	1.5
'-'	-0.52	0.7
' you'	-0.53	0.5
' can'	-0.56	0.5

Figure 5: **The cancellation is carried by a few high-frequency rows.** *Left:* Lorenz curve of the per-row gradient mass ($\|g_i^{\mathrm{ce}}\|\|g_i^{\mathrm{mtp}}\|$, ranked descending); 47 rows account for two-thirds of all mass (dashed line), against the line of equality (dotted). *Right:* the highest-mass opposed rows, decoded, are the most frequent function words and punctuation; the top four carry $\approx$41% of the total mass. Muon, seed 42, step 1000.

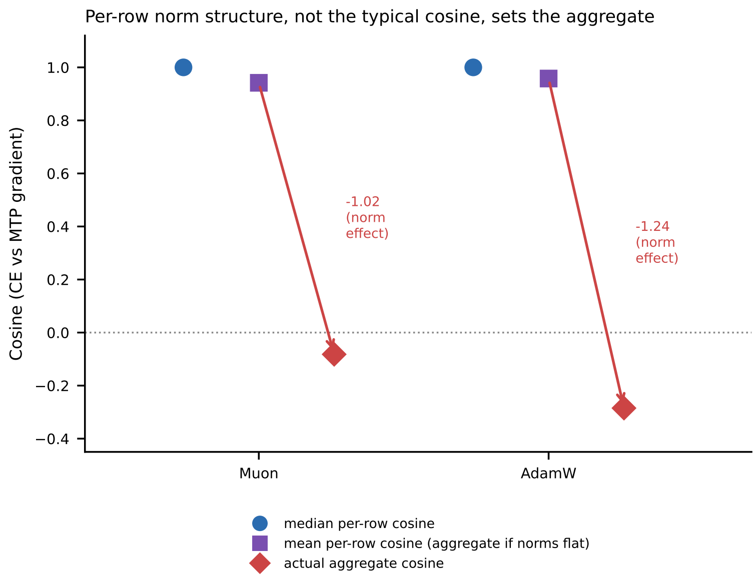


Figure 6: **Per-row norm structure, not the typical cosine, sets the aggregate.** For each optimizer: the median per-row cosine (+1.00), the plain *mean* per-row cosine (which equals the aggregate the standard diagnostic would report if gradient magnitude were spread evenly across rows), and the *actual* aggregate cosine. The mean is strongly positive (+0.94/+0.96) while the actual aggregate is negative (−0.08/−0.28); the gap is due entirely to per-row norm weighting. Single diagnostic run per optimizer, step 1000, 50,257 real vocabulary rows.

quantity	Muon (n=3)	AdamW (n=3)
active fraction of rows (%)	8.1 ± 0.3	8.0 ± 0.3
active-row median cosine	0.529 ± 0.008	0.547 ± 0.016
active rows opposed, by count (%)	3.7 ± 0.2	3.2 ± 0.2
norm-profile cosine, full vocabulary	0.979 ± 0.003	0.977 ± 0.007
norm-profile cosine, active rows	0.978 ± 0.004	0.976 ± 0.007
opposed-norm fraction, full vocabulary	0.604 ± 0.026	0.692 ± 0.011
opposed-norm fraction, active rows	0.608 ± 0.020	0.708 ± 0.010
aggregate cosine	-0.163 ± 0.030	-0.317 ± 0.020

Table 4: The rows carrying supervision are aligned, yet the mass sits on rows that oppose. Per-row and per-norm statistics at the output projection, measured at step 1,000 on the shared-MTP variant. Entries are across-seed means ± sample standard deviations (ddof=1) over seeds 42/123/456; per-seed values are in Appendix A. Rows are classified active by the norm-ratio proxy of Section 4.1. The first three rows are the decoupling result of Section 4.1: on rows carrying supervision the two losses are aligned, with only a few percent opposed by count. The next four are the mechanism of Section 4.3: a norm-profile cosine near 1 shows the two losses load the *same* rows, so the negative aggregate in the last row cannot be disjoint support, and an opposed-norm fraction above 0.5 shows the opposed minority carries most of the gradient mass. Both denominators are given because they are not interchangeable: the full-vocabulary column includes the ≈92% of rows that are scalar multiples by construction. Every entry is emitted by `analysis/stats.py` from the committed per-row arrays.

4.4 The pattern is general, and survives untying

High per-row alignment is **not unique to the output head**. Across all 74 weight matrices (Figure 7), attention and MLP matrices also show high per-row alignment (many above 0.8), frequently with strongly *negative* aggregate cosine (down to −0.78). This bears directly on what a practitioner's signal would say:

surgery methods compute conflict on the full flattened parameter vector or per shared tensor, not on the head alone, and at step 1000 the per-matrix aggregates have median -0.36 (Muon) and -0.34 (AdamW), with 89% and 86% of the 74 matrices negative. An all-parameter aggregate would therefore very likely read clearly negative under *both* optimizers. We cannot state it exactly: the snapshots committed the per-matrix cosine and per-row fraction but not the per-matrix gradient norms, so the norm-weighted combination across matrices is not recoverable from the released data. The near-zero reading that a Muon practitioner would see is a *head-level* reading, which is the level at which the loss is paid, and the false-negative arm should be read with that scope. The output head is distinctive in two ways. It is the point of **maximum** per-row alignment (0.984) combined with a near-zero aggregate; seven attention matrices sit closer to zero, but all of them at far lower per-row alignment. More importantly, it is the matrix that *directly produces the logits*. The head matters not because the metric singles it out, but because interference there is paid directly in next-token loss. Alignment also **persists when the head is untied** (A4: 99.3% of rows above threshold, versus 98.7% tied at the matched step, which is step 500: the untied ablation was run to 500 steps, so both figures are read there, and the 98.4% quoted above is the tied run at step 1000), so it is a property of the output-projection gradient, whose row i is $g_i = \sum_t \delta_i(t)\, h_t$ with $\delta(t) = \partial\mathcal{L}/\partial z_t$ the logit gradient at position t, not an artifact of weight tying. These two percentages are *full-vocabulary* and therefore construction-floored: the construction makes $\approx$92% of rows parallel, and a thresholded count on runs that committed only cosines puts $\geq$75% of all rows in that class. The count is a floor rather than an estimate, because a cosine stored in float32 need not read exactly $+1$ for a row that is parallel by construction (Section 4.8). The untied run did not commit per-row norms, so we cannot restate these figures on the active-row denominator (rows receiving a target in the batch; Section 4.1); we report them only as a point check that untying does not *reduce* alignment. The evidentiary weight of this check is modest in any case: the soft-label identity (Section 4.5) predicts untied-persistence *a priori*, since that identity never references tying. The per-layer generality (Figure 7) and the untied-head comparison are each *single-run* measurements at a fixed step; we report them as point observations, not seed-replicated estimates (only the 124M CE-vs-MTP loss gap is replicated across seeds).

We are deliberate about what this generality does and does not license. The same qualitative signature (high per-row alignment with a non-positive aggregate) appears across the network, and at many matrices the aggregate is *strongly* negative, not merely near zero. This means the per-row/aggregate gap is a fairly generic property of these structured gradient matrices, so a near-zero aggregate on its own is *not* a specific signature of two-loss interference. What is specific to the output head is that its per-row gradients are the CE and MTP gradients directly. It is the one matrix whose norm-weighted opposition we measure end to end (Figure 4), and whose interference is paid immediately in next-token loss. We do not adjudicate the per-row norm structure of the strongly-negative aggregates elsewhere in the network; a per-row account of those is beyond what our measurements settle.

A second control sharpens what the active-row alignment means, and it cuts both ways. Two next-token losses computed on disjoint halves of the same batch, through the same head, read an active-row median cosine of -0.23 to -0.24 (-0.236 on the CE-only snapshot, -0.231 on the MTP-trained one; a row counts as active here if it receives a $t{+}1$, $t{+}2$ or $t{+}3$ target in either half, since the 0.75-ratio test of Section 4.1 requires an MTP gradient to form a ratio against). The same loss on different data is thus mildly *anti*-aligned row by row wherever supervision lands, because the target-indicator component of the gradient is data-specific. On the full vocabulary the same measurement reads a median of $+0.71$, dominated by the inactive rows, where the shared softmax-push term aligns any two CE-like gradients regardless of data. Two disjoint-data next-token losses consequently read 91–93% above the $|\cos| > 0.3$ threshold, against 98.4% for CE-vs-MTP. The full-vocabulary statistic is therefore close to saturated for any pair of CE-like losses through this head and carries little pair-specific information, which is the empirical case for the active-row denominator in one number. The $+0.53$ CE-vs-MTP active median is accordingly not generic head geometry: it is specific to two losses sharing the batch, the logits, and (up to offset) the target set. The measurement decomposes the identity of Section 4.5, which rewrites the shared-head objective as one cross-entropy against a blended next-and-future target, into three parts: the push term aligns everywhere, shared targets align the active rows, and disjoint targets anti-align them.

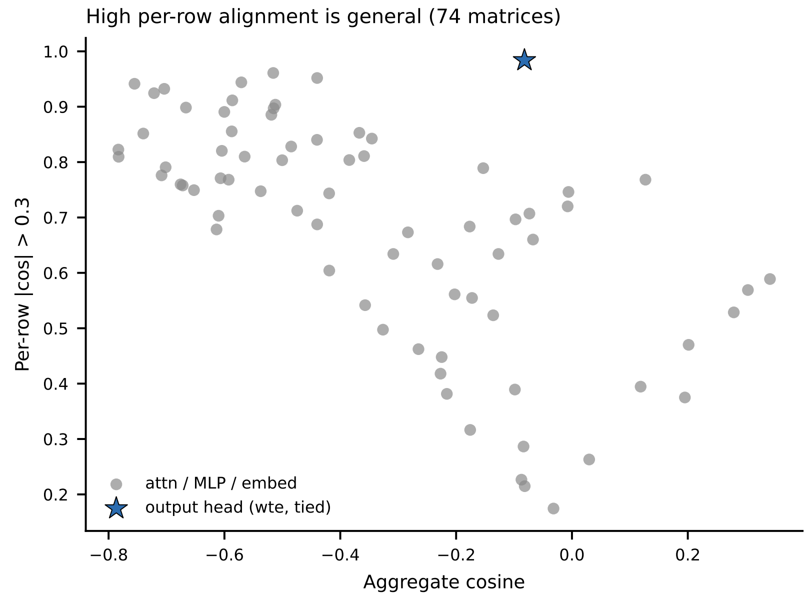


Figure 7: **High per-row alignment is general across the network.** Per-row alignment vs aggregate cosine for all 74 weight matrices. The output head (star) is the per-row maximum and combines that with a near-zero aggregate (seven attention matrices sit closer to zero, all with far lower per-row alignment), but attention/MLP matrices show the same pattern, many with strongly negative aggregate cosine. The vertical axis is the unsigned fraction $|\cos| > 0.3$, which counts an anti-aligned row and an aligned one alike; for the strongly-negative matrices on the left it therefore measures how far per-row cosines sit from zero, not how many rows agree. The signed decomposition is available only for the output head (Figures 4 and 6), where per-row cosines and norms were logged; the earlier per-layer snapshots store only this unsigned fraction, so we do not adjudicate the sign structure elsewhere in the network. Single Muon run (`results/phase_a/a1_muon_interference.json`), snapshot at step 1,000.

4.5 What the shared-head objective optimizes

Because the next-token loss and both MTP terms read the same logits z, the total objective admits an exact rewrite. Writing each cross-entropy as $\mathrm{CE}(z, y) = \mathrm{LSE}(z) - z_y$,

$$\mathcal{L} = \mathrm{CE}(z, y_1) + \tfrac{1}{2}\mathrm{CE}(z, y_2) + \tfrac{1}{4}\mathrm{CE}(z, y_3) = 1.75\,[\mathrm{LSE}(z) - \langle \tilde{q}, z\rangle]\,, \qquad \tilde{q} = \frac{e_{y_1} + \frac{1}{2}e_{y_2} + \frac{1}{4}e_{y_3}}{1.75}. \tag{2}$$

The shared-head objective is identically 1.75× cross-entropy against a soft mixture label placing 57% of its weight on the true next token and 43% on future tokens: label smoothing toward the future. Two consequences follow.

First, the per-row structure of Sections 4.1 and 4.3 is the gradient signature of this soft label. Rows never touched by $\tilde{q}$ are exact scalar multiples of one another (the 0.75 identity), and the high-mass opposed rows are precisely the frequent tokens on which the next-token target and the future components of $\tilde{q}$ disagree (Figure 5).

Second, the objective's *optimum moves.* The Bayes-optimal prediction under Equation 2 is the mixture $q^*(\cdot \mid \mathrm{ctx}) = \left(P_1 + \frac{1}{2}P_2 + \frac{1}{4}P_3\right)/1.75$, the $s{=}1$ case of the weighted family $q_s = \left(P_1 + \frac{s}{2}P_2 + \frac{s}{4}P_3\right)/(1{+}0.75s)$ used for the sweep below. Note the distinction from $\tilde{q}$ in Eq. (2). There, $\tilde{q}$ is the random one-hot label mixture on a single position; here, q^* and q_s are the corresponding mixtures of the true conditional *distributions*, with P_j the true distribution of the token at offset j. The excess next-token loss incurred at that optimum is $\mathrm{KL}(P_1 \| q^*) \le \log 1.75 = 0.560$ nats. The observed +0.39-nat degradation (Section 4.7, Table 1) is 70% of this ceiling, which raises the question the next subsection tests directly: how much of the degradation is the predictable cost of optimizing a mixture target, rather than an optimization pathology.

4.6 A weight sweep and a zero-parameter KL prediction

Weight sweep. Following the protocol of Section 3.5, we retrain at auxiliary scales $s \in \{0, 0.25, 0.5, 1.0\}$. The endpoints reproduce the committed conditions, so the sweep is anchored. We compare against the matched seed-42 runs throughout, since the sweep is itself a single seed. Shared-MTP@42 is 4.3994, displayed as 4.399 in Table 5 of Appendix A. CE-only@42 is 4.0028, displayed as 4.003 in that table's caption; the table body carries the five-seed CE-only mean, 3.985, not the seed-42 run. Against those the sweep reads lower at both ends, by 0.009 at s=0 (3.994) and 0.034 at s=1 (4.366; the unrounded operands give 0.0338, so the displayed subtraction differs by 0.001). We attribute this to bf16 nondeterminism, independent batch ordering and a separate driver script. The s=0 miss sits well inside the $\sigma = 0.0255$ run-noise scale used throughout and the s=1 miss marginally outside it, so we read the anchor as a consistency check at single-run resolution rather than a tight calibration. Neither endpoint reuses a committed run: all four sweep cells were trained by the sweep driver. The within-sweep gaps relative to its own s=0 run are 0.091, 0.183 and 0.372 at $s = 0.25, 0.5, 1.0$: proportional to s to within run noise. Under the mixture account the predicted curve $\mathrm{KL}(P_1 \| q_s)$ is itself near-linear over this range: normalized to its own s=1 value it spans 0.243–0.267 at s=0.25 and 0.511–0.538 at s=0.5 across all estimator variants, against exactly-linear 0.25 and 0.50. Rescaled to the measured s=1 gap those shape differences, measured as the largest departure from exact linearity, are at most 0.006 and 0.014 nats, i.e. 0.25σ and 0.56σ, so the sweep's shape cannot discriminate the mixture account from an interference cost proportional to the auxiliary weight. We report it as an anchored proportionality check, which it passes, and rest the discrimination on the absolute prediction below.

Comparing a KL computed at the CE-only optimum against a gap measured between two trained runs assumes both runs sit comparably far from their own objectives' optima at 4 tokens per parameter. The entropy and future-offset checks below show the shared-MTP model has actually moved toward the mixture rather than merely trained less well. They are our evidence for reading the gap as dominated by the cost of the target rather than by optimization speed.

Zero-parameter KL prediction. Using only the CE-only (s=0) checkpoint, we estimate $\hat{P}_1$ and $\hat{P}_2$ (offset 3 is bracketed rather than estimated, Section 3.5) on 2,048 held-out positions per validation half (4,096 across the split), reaching coverage 0.90–0.92 at the largest k, and compute $\mathrm{KL}(\hat{P}_1 \| \hat{q}_s)$ with no fitted parameters at every s (Section 3.5).

Against the within-sweep gaps the prediction matches at s=0.25 (0.0914 observed against band top 0.0897) and s=0.5 (0.1832 against 0.1809): both measurements sit just above the band's upper edge, by 0.07–0.09σ of run noise, which we read as agreement at these weights rather than as a resolved miss. It undershoots at s=1 (Figure 8), where the bracketing band is $[0.2316, 0.3363]$ against 0.3715 observed: a residual of +0.035, or 1.4σ for the run-noise scale $\sigma = 0.0255$. That value is $\sqrt{2} \times 0.018$, where 0.018 is a single representative per-run standard deviation fixed in the analysis code, chosen to sit inside the range our six 124M conditions actually span (0.010–0.021). The $\sqrt{2}$ converts a per-run sd into the noise on a difference between two single runs. It is a round stand-in rather than a quantity estimated from the two specific conditions, so as a check we also computed a version that is estimated from them. Combining the across-seed sds of the two conditions being differenced (CE-only 0.0138 at n=5, shared-MTP 0.0209 at n=3) in quadrature yields 0.0251 from the unrounded per-seed sds; the four-decimal values quoted here give 0.0250. Either is essentially identical to the value we use. Residuals below are 1.4σ and 1.6–1.8σ under either (1.40 and 1.63–1.82 at 0.0251). Both figures are *run-to-run* scales: they quantify how much a retrained pair of models moves, and exclude the sampling error of the KL estimate itself, which is computed on 4,096 held-out positions. The half-to-half spread of the coverage-extrapolated estimate (0.3306 against 0.3259) is the only direct visibility we have on that second component, and we have not combined the two. The σ-multiples below should therefore be read as an upper bound on the residual's significance, not as a test. Note that the larger σ is the one that makes our residual look smaller, so this choice errs in our favour rather than against it; we report both for that reason. Truncation is an *upward* bias on the estimate, and the prediction falls linearly in uncovered mass (slopes 0.0729 and 0.0676 across the two k intervals on one validation half, 0.0670 and 0.0628 on the other), so extrapolating to full coverage gives 0.3306 and 0.3259 on the two halves: residuals of +0.041 and +0.046, or 1.6–1.8σ. We quote the range across halves rather than the more favorable one. The remaining known bias, proxy smoothing from an undertrained model, is downward and unquantified: a uniform ≈10% underestimate of that kind reconciles all three scales within noise, as would a genuine non-mixture component concentrated at full weight. Our estimator cannot separate these, so we name the residual rather than assign it.

Two independent checks agree beyond direction. The shared-MTP model's predictive entropy is 1.26 nats above CE-only's, against a truncation-biased-low prediction of +0.95 from $\hat{q}^*$, and it is far better at the future offsets themselves (t+2 cross-entropy better by 2.91 nats, t+3 by 2.57). We therefore conclude that the mixture cost is the leading candidate for the bulk of the degradation, since it reproduces the gap's scaling and, at the lower auxiliary weights, its magnitude, while a residual of ≈0.04 nats at full weight (0.055–0.066 nats measured against the multi-seed headline gap) remains open. We do not claim the account is confirmed.

4.7 Adding shared MTP degrades next-token loss; no auxiliary demonstrably recovers it

This subsection reports the degradation and the interventions that fail to remove it; Sections 4.5 and 4.6 give the account of why most of it is expected and why the interventions cannot help. Adding the shared MTP objective degrades next-token validation loss from $\mathbf{3.985 \pm 0.014}$ (CE-only, n=5) to $\mathbf{4.377 \pm 0.021}$ (n=3): a +0.39 **nat** (≈10%) increase. On the three seeds the two conditions share $(42, 123, 456)$, a paired test gives a mean gap of **0.382** nats, t=50.0, $\mathbf{p \approx 4.0 \times 10^{-4}}$ (df=2). The per-seed gaps are 0.397/0.380/0.370 as displayed; the test uses the unrounded values 0.396562/0.380312/0.370313, and recomputing from the three-decimal figures instead gives t=48.5. Treating the conditions as independent (Welch, CE-only minus shared-MTP, using all n=5 and n=3 seeds) gives $t = -28.8$ (negative because CE-only is the lower loss), $p \approx 8 \times 10^{-5}$ with a 95% CI on the difference of $[0.35, 0.43]$ nats. The gap is large relative to the seed-to-seed spread (within-condition sd ≈ 0.01–0.02), which is what makes it detectable at these small n. We report the effect as the 0.39-nat loss increase itself rather than a standardized effect size, since the latter is inflated by the tight reproducibility of a controlled testbed. With n=3 in the smaller arm and this spread, the minimum difference we have ≈80% power to detect is $\Delta \approx 0.045$ nats under the *paired* test (SE ≈ 0.008, df=2, two-sided α=0.05); at that SE, Δ=0.05 is detectable at ≈86% power and Δ=0.04 at only ≈72%. The unpaired Welch comparison is the weaker instrument: at SE ≈ 0.014 and df ≈ 3.1, Δ=0.05 carries only ≈68% power, and ≈80% needs $\Delta \approx 0.06$. We quote the paired floor because seed-matched pairing is available for these comparisons. The headline gap is nearly 9× that floor, so it is detected with room to spare, whereas differences below ≈0.05–0.06 nats (relevant to the tuned auxiliary below) are at or under our detection limit. We report uncorrected p-values for the k=5 variant-vs-baseline comparisons; under a Holm correction at α=0.05, which controls the family-wise error rate across the five tests, the shared-MTP, stop-gradient, standard-contrastive, and NextLat results remain significant, while only the tuned-auxiliary comparison ($p \approx 0.19$) does not survive, consistent with its effect sitting below the detection floor. Detaching the auxiliary from the head via stop-gradient makes it *worse* (4.493, Δ+0.508, $p < 10^{-4}$), not better. What this rules out is the naive head-detach fix: if the head's opposed rows were the site of the damage, removing the auxiliary gradient there should have recovered the baseline, and it does the opposite. It does not by itself establish which account is right, because the probe changes two things at once (see below). The probe forms the auxiliary logits as $\mathrm{sg}(W)h$ with the auxiliary weights unchanged at 0.5 and 0.25, so the auxiliary still trains the trunk and only the head's auxiliary gradient is removed. It is a data point rather than a proof: with the head frozen against the auxiliary, the head can no longer co-adapt to the representations the auxiliary shapes, which is a competing reason the run could be worse. The n=5 CE-only and n=3 shared-MTP conditions share three seeds $(42, 123, 456)$; the two extra CE-only seeds $(789, 1337)$ were run for a tighter baseline and no MTP seeds were dropped. The standard capacity-separated auxiliary reduces but does **not** recover the baseline; the tuned variant is inconclusive. The *standard* layer-split contrastive auxiliary (G_{nce}: an InfoNCE-style (van den Oord et al., 2018) contrastive loss applied to an intermediate layer, giving the auxiliary its own representation capacity instead of sharing the output head) reaches 4.024 (Δ+0.039, $p \approx 0.010$), statistically distinguishable from A, i.e. a genuine non-recovery. A hyperparameter-tuned variant reaches 4.003 (Δ+0.018) but is *not* statistically separable from A at n=3 ($p \approx 0.19$): for this variant we can claim neither recovery nor non-recovery, and we rest the capacity conclusion on the standard variant. A latent-prediction baseline (NextLat, after Teoh et al. (2025): the auxiliary predicts the *hidden*

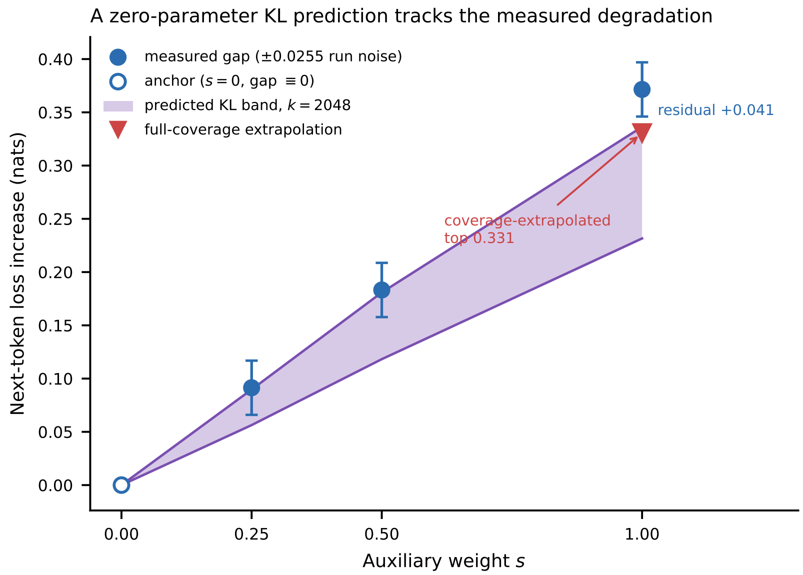


Figure 8: **A zero-parameter KL prediction tracks the measured degradation.** Next-token loss increase against auxiliary weight s, all from seed 42. Filled points are the measured within-sweep gaps with ± 0.0255 run noise; the open point at $s{=}0$ is the anchor, whose gap is identically zero by construction and therefore carries no interval. Every gap is taken against that same $s{=}0$ run, so the intervals share a common endpoint and are correlated across s rather than independent. The shaded band is the predicted $\mathrm{KL}(\hat{P}_1 \| \hat{q}_s)$ at $k{=}2048$, spanning the two offset-3 treatments and both validation halves, and it has *no fitted parameters*: it is computed entirely from the CE-only checkpoint's own distributions. The band is *designed* to bracket the true target: dropping offset 3 under-weights the future mass, while $P_3 \approx P_2$ substitutes a more peaked component. We do not prove the second direction, and one piece of our own data argues it runs the other way. Substituting a peaked component for P_3 raises $\mathrm{KL}(P_1 \| q_s)$ only if that component's mass sits away from P_1's; a peaked component overlapping P_1 lowers it. The committed sweep runs put cross-entropy against offset-3 targets 0.58–0.92 nats above offset-2 at every auxiliary weight, so P_3 is the flatter component and P_2 the one closer to P_1. Replacing P_3 by P_2 therefore pulls q_s *toward* P_1 and plausibly *lowers* the predicted KL. If that reading holds, both arms bias downward, the interval does not bracket the truth from above, and part of the $+0.035$ residual at $s{=}1$ is a construction artifact rather than a miss. We flag this rather than resolve it: settling it needs P_3 estimated directly, which we list as the highest-value remaining measurement in *Future work*. Read as written, the upper edge is a construction we intend as conservative and have not bounded. At $s{=}0.25$ and $s{=}0.5$ the measured gaps sit on the band's upper edge (above it by 0.07–0.09σ of run noise, i.e. agreement at our resolution); at full weight the measurement sits clearly above it, leaving the residual of $+0.035$ raw ($+0.041$ to $+0.046$ against the coverage-extrapolated tops 0.3306 and 0.3259 from the two validation halves; the figure marks the higher top, 0.3306, which is the smaller of the two residuals), i.e. 1.4–1.8σ for the run-noise scale $\sigma = 0.0255$. Truncation biases the estimate upward, so extrapolating to full coverage lowers the prediction and widens this residual, which we name rather than assign (Section 4.6).

state at the next offset rather than its token, so it never touches the output head) lands at 4.037 ($\Delta{+}0.051$ from unrounded means; the displayed operands subtract to 0.052, Welch $p{\approx}0.018$) (Figure 9).

Conflict-targeted gradient surgery also does not recover the baseline, and this is a stronger statement than "there was nothing to remove." The opposed rows are negligible by count (0.3%) but carry the majority of the gradient mass (60–69%), and a projection acts on the mass-weighted vector, so surgery had a great deal to project away. Per-row Gram–Schmidt on the head targets exactly those opposed rows and still recovers nothing. The measured opposition is mass-dominant and causally inert. Section 4.5 supplies the reason this should be expected. The shared objective's minimizer has moved to a mixture target, and reprojecting gradients does not move an optimum back (exact for symmetric projection, empirical for the head-only

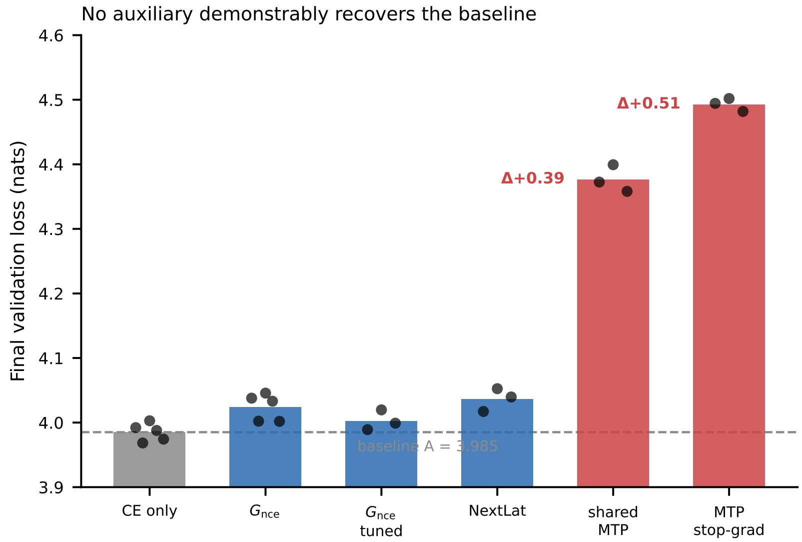


Figure 9: **No auxiliary demonstrably recovers the baseline.** Final next-token validation loss by variant (points = seeds). Shared MTP degrades CE by Δ+0.39; stop-gradient by Δ+0.51. The *standard* capacity-separated auxiliary does not recover the CE-only baseline (Δ+0.039, $p \approx 0.010$); the tuned variant (Δ+0.018, $p \approx 0.19$) is statistically inconclusive at our detection floor.

variants used here; Section 5). Because those runs are single-seed with a numerical-drift caveat, we report them in full in Appendix A rather than resting any claim on them.

4.8 The apparent rare-token alignment is a construction artifact; the pattern holds at 350M

Measured over the full vocabulary, per-row alignment appears *higher* for rarer tokens: the rarest quintile averages 0.98 mean $|\cos|$ versus 0.86 for the most-common quintile (Figure 10, left). This effect is an *artifact of the construction floor*. Rare tokens are almost never targets in a measurement batch, so they are overwhelmingly parallel-by-construction ($\cos = +1$): the active fraction, meaning the share of rows that do receive a target (Section 4.1), falls from ≈25% in the common quintile to ≈1.5% in the rarest. Restricted to active rows, the apparent gradient disappears: active-row mean $|\cos|$ is nearly flat across frequency quintiles, with a weak residual (0.51–0.54 for Muon, 0.52–0.60 for AdamW; Figure 10, right). GPT-2 BPE id is itself only a rough frequency proxy: Spearman $\rho \approx 0.12$ between token id and active-row $|\cos|$, small but nonzero at $n \approx 3{,}900$ active rows. We therefore *withdraw* the rare-token-alignment reading: the full-vocabulary ordering reflects how many rows are parallel by construction, not a frequency-dependent alignment. The aligned-per-row-with-negative-aggregate picture holds qualitatively at **350M**: CE-only 3.752, shared-MTP 4.129 (Δ+0.377). That comparison is a single seed, run with the same Muon-hybrid optimizer and the same FineWeb-Edu corpus as the 124M runs, on a necessarily larger pool. 87,000 steps at 16,384 tokens/step is 1,425,408,000 tokens processed over 1.369B unique, or 1.04 epochs and ≈ 4.0 tokens/parameter at 355M. The stream is 1.5B tokens, of which the first 1.389B are used, with the same 480M–500M validation slice held out as at 124M.

The 350M runs are the paper's only *late-training* per-row measurement, logged out to step 87,000. The instrument computes the same quantity in both runs (the per-row cosine between the CE gradient and the MTP gradient on the output projection) regardless of which objective the model was trained with (the MTP gradient is computed at measurement time for both). Measured this way, the shared-MTP-trained model holds $|\cos| > 0.3$ at 98.9% of rows at step 87,000, and the CE-only-trained model reads 94.5% at the same step. These are *full-vocabulary* figures and thus construction-floored. The two bounds we quote differ by what each run committed: runs that logged only cosines admit the weaker ≥75% parallel bound (a floor, not an estimate: floating-point noise on tiny-norm rows pushes some exactly-parallel rows below the

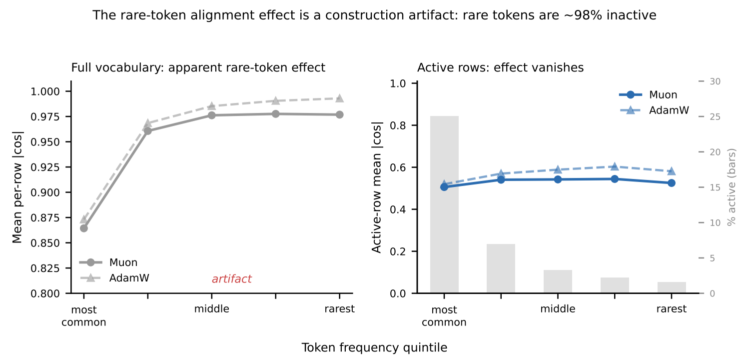


Figure 10: **The rare-token alignment effect is a construction artifact.** Left: over the full vocabulary, mean per-row $|\cos|$ rises from common to rare tokens (both optimizers); this is the apparent effect. Right: restricted to active rows (those carrying a target), the apparent gradient disappears and mean $|\cos|$ is nearly flat across frequency quintiles, with only a weak residual; the grey bars show the active fraction collapsing from ≈25% (common) to ≈1.5% (rare), which is what drives the full-vocabulary trend. Both panels plot the *unsigned* $|\cos|$, which is not comparable with the *signed* active-row fraction of ≈0.83 quoted in Section 4.1: that figure is signed and on the active-row denominator, whereas both panels here are unsigned, so the levels must not be read against each other in either direction. Single run per optimizer, step 1000, GPT-2 124M.

$1 - 10^{-6}$ cosine threshold, so the thresholded count understates the true parallel fraction), whereas runs that also committed per-row norms permit the ratio classification of Section 4.1 and give ≈92% parallel. That 92% is measured on the 124M step-1000 runs, not on these 350M runs, which committed only summary statistics; applying it here takes the 124M active fraction as representative at triple the capacity and 87× the steps, which we cannot check from the saved data. Taking the tighter figure, the informative range is the remaining ≈8 percentage points, within which the MTP-trained model sits at ≈6.9 points above the floor and the CE-only model at ≈2.5, so the 98.9-versus-94.5 separation spans over half of that range, rather than the small slice of the 25 points the weaker bound would leave (coarse, but not vacuous), stated here at its true resolution (the 350M runs committed only summary statistics, so an active-row restatement is not recoverable from the saved data). That the CE-only model *also* shows high CE/MTP alignment this late indicates a substantial intrinsic component in the output-projection gradient geometry, not induced by the MTP objective. Which term *dominates* depends on the denominator, and we state both. Against the full vocabulary the intrinsic reading is larger (94.5 points against the 4.4 the objective adds). Against the ≈8-point informative range above the construction floor the ordering reverses: the CE-only model contributes 2.5 points and MTP training a further 4.4. We therefore claim only that the intrinsic component is substantial, not that it is the larger of the two. Both are single seeds, so we treat the per-row persistence as illustrative (consistent with alignment not decaying over a long horizon), not established.

The 350M *loss* gap (Δ+0.377 above) is a next-token-loss number, unaffected by the active-row denominator, and it is this scale-stability of the gap (+0.3915 at 124M, +0.3775 at 350M) that supports the account of Section 4.5: $\mathrm{KL}(P_1 \| q^*)$ is a property of the data distribution, not of model capacity, so the mixture account predicts a gap that is roughly scale-invariant, whereas an optimization-pathology account has no particular reason to hold the gap fixed as capacity triples. The two scales differ by 0.014 nats (from the unrounded values above), within the seed spread at 124M. This is one single-seed point at 350M against three at 124M, so we read it as consistent with the mixture account rather than as a measured scaling law.

5 Discussion

What the measurements establish. In this regime (a single shared tied output projection, ≤350M, undertrained) the standard cosine conflict diagnostic misreads the CE/MTP interaction. On the rows that carry a target the per-row gradients are aligned (active-row median cosine ≈0.53, ≈3–4% opposed; the full-vocabulary median is +1 only because ≈92% of rows are scalar multiples by construction), the aggregate is

≈ 0 or negative, and the two decouple during training. Instrumenting the per-row norms shows the aggregate is set by a high-norm opposed minority ($\approx 0.3\%$ of all rows, carrying 60–69% of the mass), not by disjoint support. The losses load the same rows (norm-profile cosine 0.98) but oppose on the few rows that dominate the weighted sum. Crucially, the next-token degradation is large and unambiguous, yet neutralizing that directional opposition does not remove it. The surgery baselines in Appendix A recover nothing, which is consistent with the opposed minority not being the fixable lever. Under the soft-label reading this is expected for a sharper reason than opposition that carries the mass yet is causally inert: the objective's *minimizer has moved* to the mixture q^*, and reprojecting gradients does not move it back. For symmetric PCGrad the argument is exact: at a stationary point of the summed loss $g_{\mathrm{CE}} = -g_{\mathrm{MTP}}$, both project to zero, and the mixture optimum stays stationary. Our asymmetric head-only Gram–Schmidt is not a fixed point of those dynamics in the same way, so there the empirical null is the evidence, and the natural reading is that a head-only correction cannot undo a shifted optimum while the trunk keeps receiving the full mixture gradient and shaping representations toward q^*. Capacity separation helps precisely because the auxiliary no longer forces one set of logits to serve every horizon, though in our runs it reduces rather than eliminates the gap.

The stop-gradient result cuts the other way. One intervention appears to contradict the shifted-optimum account, and we state it plainly rather than leave it to a reader. If the head's blended target were the whole story, then $\mathrm{sg}(W)h$ should help: the head is freed to minimize pure next-token cross-entropy given the representation it is handed. Instead stop-gradient is the worst of the auxiliary conditions in Table 1 (4.493, Δ+0.508 against shared MTP's Δ+0.392); only the gradient-surgery and optimizer-coverage variants of Appendix A sit higher. The reason we cannot read this as a refutation is that the probe moves two quantities at once. It removes the head's mixture target, which the account predicts should help, and it also stops the head from co-adapting to a trunk that is still being shaped by the full mixture gradient, which should hurt. The observed sign says the second effect is the larger one; it does not say the first is absent. Separating them needs an intervention that removes the mixture target from the head *without* decoupling head and trunk, which is what the separate-head variant in *Future work* does. Until then we treat the stop-gradient number as evidence against the naive head-detach fix and as uninformative about the mixture account itself.

Beyond the diagnostic result, an exact rewrite shows the shared-head objective is soft-label cross-entropy toward a next-and-future-token mixture (Section 4.5), and a zero-parameter estimate of the implied KL cost, computed entirely from the CE-only model's own distributions, matches the observed degradation within run noise at the lower auxiliary weights while the full-mixture variant reproduces ≈88–89% of the matched within-sweep gap at full weight, leaving a ≈0.04-nat residual attributable to the estimator's remaining downward proxy bias and/or a small genuine non-mixture component (Section 4.6). The anchored weight sweep shows the degradation scales with the auxiliary weight as this account predicts, as a weight-proportional interference account would also predict; the two shapes are indistinguishable at our noise level.

Two failure modes, one instrument. The aggregate cosine misleads in two opposite directions depending on the optimizer, and the paper's central claim is that both are failures of the *same* diagnostic. Under Muon the aggregate is -0.08 to -0.19 across measurement batches, close enough to zero at its shallowest reading (-0.08) that a conflict diagnostic reads "no interference," a *false negative*: the next-token loss degrades by 0.39 nats regardless. Under AdamW the aggregate is -0.28 (-0.30 to -0.34 over the norm-committing seeds), negative enough to trip a conflict flag, and the surgery runs are *consistent with* a false positive: projecting out exactly this opposition recovers nothing (Appendix A). Those runs are single-seed, with a drift envelope (0.02–0.04 nats) that overlaps the effects they would need to resolve, so this arm is suggestive rather than established; the matched-seed replication that would settle it is future item 3 below. A practitioner reading only the head's aggregate would conclude "safe to add MTP" under Muon and "apply surgery" under AdamW, and both conclusions are wrong, the first demonstrably and the second on suggestive evidence. The false negative is specific to the head: as Section 4.4 notes, an all-parameter aggregate would very likely read clearly negative under both optimizers, so the misread is of *where* interference lives rather than of whether any is present. The instrument does not merely lose power; it points the wrong way, and which way depends on an optimizer choice unrelated to whether interference is present. This is why the per-row and per-norm decomposition, not a scalar threshold on the aggregate, is what the regime requires.

6 Limitations

We state the boundaries of the result under four headings: what the scope of the claim is, what is not established about mechanism, what biases the estimator carries, and where the statistics are thin.

Scope. Every measurement is on a single shared tied output projection at ≤350M parameters, in an undertrained, small-batch regime where MTP is already documented to underperform (Gloeckle et al., 2024; Zuhri et al., 2025). We do not show that MTP *as practiced* (with separate per-horizon heads) degrades next-token modeling. The reason separate heads should help is not that they stop the losses from sharing rows: the standard designs share one unembedding matrix, so per-row gradient mixing persists there. It is that per-horizon logits remove the constraint that one distribution serve every horizon, which is precisely the optimum shift of Equation 2. That makes the separate-head experiment a clean test of our own account rather than a mere scaling check, since it holds shared rows fixed and varies only the shared-label ingredient. The outcome measure is also narrow: we record next-token validation loss, not downstream task performance, so "degrades" should be read as that quantity and no more.

Mechanism. We do not establish causation between the norm-weighted opposition and the loss increase. We show they co-occur, and that interventions behave as a shifted-optimum account predicts: surgery that removes the opposition does not remove the degradation, and separating capacity helps. A related gap is one of timing rather than of inference. The per-row and per-norm measurements are taken early in training, at step 1,000, while the loss comparisons are at 30,517 steps; late-training persistence of the alignment is measured directly only at 350M, on a single seed. The aligned-gradient state and the degraded-loss state are therefore never exhibited in the same measurement at 124M, and the connection between them rests on that persistence rather than on a co-located observation.

Estimator. The zero-parameter prediction sits below the observed gap by 0.035–0.046 nats: within its stated biases, but also within what a small genuine interference residual would produce. We treat the mixture account as the leading candidate, not as confirmed, and we name the residual rather than assign it. The estimate is also truncated rather than fully marginalized, and truncation biases it upward relative to full marginalization, so a fuller estimate would lower the prediction and *widen* the residual rather than close it.

Which gradient we measure. The head is weight-tied, so the gradient our instrument reads sums the output-projection path and the embedding path (Section 3.2). The embedding path deposits only on rows that already carry a target, so it lands exactly on the active rows every statistic here is computed over, and we do not bound its share. Two measurements would close this, and neither needs new training. One is to recompute the projection path alone, taking the gradient with respect to the logits and forming $\sum_t \delta_i(t) h_t$ directly. The other is to commit per-row norms from an untied run, where the head tensor receives the projection path only. The existing untied run cannot serve, because it committed full-vocabulary fractions rather than per-row arrays.

Statistics and instrument caveats. Seed counts are small and uneven: n=5 for CE-only and the standard G_{nce} auxiliary, n=3 for the other four conditions (Table 1). The auxiliary-weight sweep of Section 4.6 is a single seed throughout, so the strongest quantitative claim in the paper carries no across-seed spread of its own. The untied-head check is one run at 500 steps against 30,517 elsewhere. Finally, the full-vocabulary form of the CE-vs-L1 control is threshold-sensitive and we do not rest calibration on it. On the committed diagnostic run the fraction of rows with $|\cos|$ above 0.1, 0.2, 0.3 and 0.5 is 74.9%, 35.2%, 0.42% and 0: nearly 35% of rows fall between the 0.2 and 0.3 bins, so the above-0.3 figure is sensitive to exactly where the cut is placed, and it reads 36.2% on the sweep snapshot. That snapshot committed only these binned fractions, not the underlying per-row cosines. We therefore cannot say whether the two runs differ by a small shift near this quantile or more substantially, and we do not claim the discrepancy is explained. The shape of that distribution is itself unexplained, and we flag it rather than account for it. Two comparisons bound the problem. Against isotropic vectors in d=768 the cosine standard deviation is 0.036, so almost no rows should clear 0.1, yet 74.9% do: the rows are far more aligned than chance. But the profile is also *sharper* than any low-dimensional-span model predicts. Suppose both gradients were random within a shared k-dimensional subspace. Any k that reproduces the above-0.1 fraction (within 5 points of the observed 74.9%) forces the ratio of the above-0.2 to the above-0.3 fraction to at most $\approx$2. The observed ratio is 84. Larger k can produce a sharp ratio, but only by putting far too little mass above 0.1. The best-fitting single k is 23, and it still overpredicts the above-0.3 fraction by a factor of $\approx$37. So neither a chance model nor a shared-span model reproduces the cliff, and we do not have per-row cosines from this run to diagnose it. The control's role is to show that an *unrelated* loss pair yields a near-zero aggregate (0.002), not to quantify a per-row distribution. The unexplained shape therefore does not bear on the calibration claim. It remains a loose end: committing per-row cosines for the control is the measurement that would close it. What both runs agree on is the property the control exists to establish: the aggregate cosine is near zero either way (+0.002 and −0.011). This is why the active-row control of Section 3.4 is the one we report.

Future work. Five directions would broaden the scope from "the diagnostic misreads this regime" toward a causal, general statement.

(1) A **separate-head MTP** variant, to localize whether the degradation is intrinsic to head-sharing.

(2) A **co-located measurement** at 30,517 steps plus a 124M CE-only per-row baseline, to replace the timing assumption noted under *Mechanism* above with a direct observation, and n=3 at 350M to turn the single-seed scale observation into a measured trend.

(3) **Matched-seed, single-pass gradient surgery** at n=3, to remove the numerical-drift caveat.

(4) A **tighter KL estimate** (full marginalization or larger k, and better-trained proxy models), to bound the 0.035–0.046-nat residual from the prediction side rather than the measurement side. The highest-value single piece is **estimating P_3 directly** instead of substituting P_2. That is the one bracket arm whose direction we have not bounded. Our own offset-3 cross-entropies suggest the substitution biases the prediction downward, so removing it could shrink the residual rather than merely bound it.

(5) **Committing per-row cosines for the control** at both checkpoints, which is what would settle the threshold-sensitivity question raised under *Statistics and instrument caveats* above. Both snapshots are preserved, but neither stores the per-row array.

7 Conclusion

Cosine similarity between task gradients is the default instrument for detecting interference in multi-task and auxiliary-loss training. We showed that at a language model's shared, tied output projection this instrument is misleading, in three respects. The per-row CE and MTP gradients are aligned in direction on the rows that carry a target (active-row median cosine $\approx$0.53; the full-vocabulary median is +1 because most rows are scalar multiples by construction). The aggregate cosine is driven negative by a high-norm opposed minority ($\approx$ 0.3% of rows carrying most of the mass), not by a broad directional conflict. And the two metrics decouple during training. Adding the shared auxiliary degrades next-token loss substantially, and the surgery baselines we ran, single-seed and reported in Appendix A, do not repair it. The aggregate is set by per-row gradient norms, not by the per-row directions the cosine averages, a distinction that explains why conflict-targeted surgery does not help here and why capacity separation does. An exact rewrite of the objective supplies the complementary positive account: the shared head optimizes cross-entropy against a next-and-future-token mixture, so its optimum has moved, and a zero-parameter estimate of the resulting KL cost reproduces most of the measured degradation, undershooting it by 0.035–0.046 nats at full auxiliary weight. On this reading the degradation is a predictable consequence of the objective rather than a pathology of optimization. Gradient surgery is then inert: exactly so for symmetric projection, and empirically so for the head-only variants we test, where the trunk continues to receive the full mixture gradient. We deliver the result on a fully reproducible testbed with a committed statistics script, and we are explicit about the shared-head, small-scale scope and the experiments that would extend it.

Reproducibility Statement

All results are computed from committed result JSONs by `analysis/stats.py`; all data figures are regenerated from the same JSONs by `figures/make_figures.py` (the conceptual schematic, Figure 1, by `figures/make_schematic.py`). Code, data-loading, pinned dependencies (`requirements.txt`), and a `README` with exact run commands are available at `[[ TODO BEFORE SUBMISSION: anonymized repository URL ]]`. The optimizer (Muon) is pinned to commit `056a3c5869cf`. Every training and measurement run reported here was performed on a single rented NVIDIA H100, booked through the Prime Intellect compute exchange on hardware operated by Verda (DataCrunch). Across the 74 committed runs the recorded wall-clock totals 196,501 seconds, or ≈54.6 H100-hours; that counts the runs whose JSONs ship here and excludes exploratory and failed attempts, which were not committed and whose time we did not log. (That 74 is coincidental and unrelated to the 74 weight matrices of Section 4.4.) This work received no external funding. Scope and caveats are stated in Section 1 and Section 6.

Use of large language models. A large language model was used as a writing and analysis assistant: drafting and revising prose, generating figure-plotting and statistics code, and cross-checking arithmetic against the committed result files. It was not used to generate data, and no experimental result, number, or citation in this paper originates from a model without being recomputed from the committed JSONs or verified against the cited source. The author is responsible for all claims.

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A Surgery and control baselines (single-seed; reported for completeness)

We ran conflict-targeted gradient-surgery methods on the shared-MTP variant to test directly whether removing directional conflict recovers the baseline. **We do not rest the paper's diagnostic result on these runs**; we read them as consistent-with rather than as evidence, for three reasons stated up front.

(i) Each is a **single seed** (seed 42).

(ii) The PCGrad baseline operates on **all parameters**, whereas per-row Gram–Schmidt and Scatter operate on the **output head only**, so they are not strictly comparable.

(iii) Surgery uses **two backward passes** under bf16, versus the fused single pass of the main runs, introducing a ∼0.02–0.04 nat numerical drift comparable to the smaller surgery effects.

With those caveats, the pattern is uniform: no surgery method recovers the CE-only baseline (A = 3.985), and none improves on the matched shared-MTP seed (B@42 = 4.399). Two of them make it measurably worse. Per-row Gram–Schmidt at +0.018 sits inside the 0.02–0.04 nat drift envelope, so we read it as no recovery. Scatter at +0.058 and PCGrad at +0.145 are outside that envelope, and PCGrad and Pure-Muon also act on all parameters rather than the head alone. We report the worsening as measured and do not explain it: removing the opposed component discards gradient magnitude that the losses agree on, but we have not isolated that from the optimizer coverage difference, so the direction is established and the cause is not.

The three surgery methods differ in what they do to the conflicting component. PCGrad (Yu et al., 2020) projects each task gradient onto the normal plane of the other whenever their inner product is negative. Per-row Gram–Schmidt applies the same idea row by row at the head, removing from the MTP row-gradient its component along the CE row-gradient wherever the two oppose. Scatter instead zeroes the MTP gradient on every opposed row and rescales the surviving rows so that the total MTP gradient norm is preserved, redistributing the removed magnitude onto the aligned rows rather than discarding it.

The uniform non-recovery is what a norm-weighting/capacity account predicts. Under the soft-label reading of Section 4.5 it has a sharper explanation still: the objective's minimizer has moved, and reprojecting gradients does not move it back (exact for symmetric projection, empirical for the asymmetric head-only variants tested here; Section 5). It is stronger than "nothing to project away." The opposed rows carry the majority of the gradient-magnitude mass (60–69%; Section 4.3), and a projection acts on that mass-weighted vector, so there was substantial opposition to remove; per-row Gram–Schmidt on the head targets exactly those rows and still leaves the loss at the shared-MTP baseline. The measured opposition is real and high-magnitude, yet causally inert. A matched-seed, single-pass replication at n=3 (identified in Section 6) would remove the drift caveat and is the right way to upgrade this from a consistent-with observation to evidence. A second-optimizer replication (AdamW; `results/phase_c/`) reproduces the *ordering* at the same scale and budget: CE-only = 3.743/3.754, shared MTP = 4.149/4.153, G_{nce} = 3.778/3.781 across seeds 42/123, so shared MTP again degrades CE-only (here by ≈0.40 nats) and the capacity-separated auxiliary again lands

Method	Acts on	Final val loss	Δ vs B@42	Δ vs A
CE only (A, ref)	—	3.985	—	0.000
shared MTP (B@42)	—	4.399	0.000	+0.414
Per-row Gram–Schmidt	head only	4.418	+0.018	+0.432
Scatter	head only	4.458	+0.058	+0.472
PCGrad	all params	4.544	+0.145	+0.559
Pure-Muon	all params	4.556	+0.157	+0.571

Table 5: No surgery variant recovers the CE-only baseline. Surgery and control baselines on shared MTP (seed 42). "Pure-Muon" places every 2D parameter under Muon, including the embeddings and output head that the hybrid recipe assigns to Adam, so it is an optimizer-coverage variant rather than a surgery method. Every method leaves the loss at or above the shared-MTP baseline; the per-row Gram–Schmidt effect (+0.018) sits inside the two-pass drift envelope, so we treat it as "no recovery," not as a measured effect. Deltas are computed from unrounded losses; the displayed operands are rounded to three decimals, so a displayed subtraction can differ by 0.001. Note the two reference columns are not on the same footing: Δ vs B@42 is a matched-seed comparison, whereas Δ vs A uses the five-seed CE-only mean (3.985) against single-seed surgery runs. Against the matched CE-only seed (4.003) every Δ vs A would be 0.018 smaller, so that column reads slightly large relative to a like-for-like pairing.

between them. These runs use the same 124M model, the same 30,517 steps and 500M-token FineWeb-Edu pool, and the same 16×1024 batch as the Muon runs, but a *different optimizer recipe*: plain AdamW at 3×10^{-4} on *all* parameters, cosine-decayed to 10% of base with no weight decay, against the Muon hybrid used everywhere else (Muon on the 2D hidden weights at a *flat* lr=0.02, wd=0.01, never annealed; Adam on embeddings and head at 3×10^{-4} cosine-decayed, wd=0.1). The two arms therefore differ in learning rate, decay schedule and weight decay on 85M of 124M parameters, and the hybrid's hidden weights are still at full learning rate at the last step. We did not run the ablation that would isolate the contribution of any one of these differences. We therefore do not attribute the $\approx$0.24-nat difference from the Muon baseline (3.985) to a specific cause; the comparison is confounded by the recipe and carries no information about which optimizer is better here. Absolute losses are **not** comparable across the two recipes, and we read only the within-optimizer ordering. Corpus, tokenizer, token budget, batch shape and validation split are identical (both stream `HuggingFaceFW/fineweb-edu`, `sample-10BT`, with the final 20M tokens held out and `final_val` averaged over 50 batches). All runs in this appendix are 124M; the 350M results of Section 4.8 are separate single-seed Muon-hybrid runs.

Per-seed values behind the aggregate statistics. For the norm-support measurement (Section 4.3) the per-seed opposed-norm fractions are, for Muon, 0.6069 (seed 42), 0.6281 (123), 0.5772 (456), and for AdamW 0.6914 (42), 0.7035 (123), 0.6806 (456); the norm-profile cosines are 0.9783/0.9761/0.9820 and 0.9846/0.9713/0.9739 respectively, and the aggregate cosines $-0.1928/-0.1631/-0.1330$ and $-0.3186/-0.3355/-0.2964$. Restated on the active-row denominator (Section 4.1), the same seeds give opposed-norm fractions 0.6301/0.6028/0.5909 (Muon) and 0.7088/0.7179/0.6988 (AdamW), norm-profile cosines 0.9776/0.9734/0.9816 and 0.9843/0.9710/0.9732, and parallel-mass fractions f of 3.89%/11.52%/2.34% and 3.53%/2.82%/2.82%; the Muon seed-123 value of f is the outlier discussed in Section 4.1. The per-row direction statistics of Table 4 are, on the same seed order, active fractions of 7.82%/8.18%/8.36% (Muon) and 7.70%/8.02%/8.21% (AdamW), active-row median cosines 0.5363/0.5302/0.5207 and 0.5637/0.5434/0.5325, and active rows opposed by count 3.51%/3.91%/3.69% and 2.94%/3.38%/3.20%. The Welch comparison of the opposed-norm fraction between optimizers ($t = 5.42$, df $= 2.77$, two-tailed $p = 0.015$) is computed from the full-vocabulary values above; the same comparison on the active-row values gives $t = 7.82$.